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EVIDENCE FROM NEW YORK CITY

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Congestion Pricing and Emergency Medical Service Response: Evidence from New York City

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ABSTRACT

Large cities worldwide have adopted congestion pricing to reduce urban traffic, with well-documented benefits for travel speeds, accident rates, and air quality. This paper identifies a novel external benefit: faster emergency medical service (EMS) response times. We provide the first evidence on how the congestion pricing program in New York City, the first comprehensive cordon-based congestion pricing system in the U.S. that was implemented on January 5th 2025, affects EMS performance. Exploiting the sharp geographic boundary of New York City's congestion relief zone at 60th Street and a difference-in-discontinuities design, we first document substantial changes in traffic and mobility near the boundary: vehicular traffic declines by roughly 18 to 21 percent, accompanied by increases in pedestrian and bicycle activity. Consistent with these first-stage mechanisms, we find that congestion pricing improves EMS performance, reducing total travel times by 63–70 seconds (approximately 5–6 percent). Effects emerged quickly and show little evidence of displacement to adjacent areas. A concurrent FDNY directive requiring transport to the nearest hospital confounds standard difference-in-differences estimates but not our boundary-based design. These findings suggest that cost-benefit analyses of congestion pricing systematically understate net social benefits by omitting emergency response improvements.

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Introduction

Large cities around the world, including Singapore, Oslo, Rome, London, Stockholm, and Milan implemented congestion pricing policies to reduce urban traffic congestion. Research generally concludes that these policies effectively reduce congestion (Veitch and Rhodes, 2024). In addition to direct benefits, Anas and Lindsey (2011) identify positive externalities generated by congestion pricing beyond the primary goal of congestion relief. Pricing reduces accidents — London saw a 2–5 percent drop in personal injury accidents, Stockholm 5–9 percent, and Milan over 20 percent — and improves air quality. These three cities also experienced reductions in airborne particulate matter like NO_x, PM₁₀, and CO₂. A second type of external benefit flows to public transport users because fewer cars on the road allow buses to travel faster and more reliably. Shifts in transportation mode can be amplified when toll revenues are reinvested in transit capacity, exploiting scale economies in public transport.

We investigate a novel type of external benefit generated by congestion pricing: reductions in emergency medical service (EMS) ambulance trip times. Three recent papers highlight the importance of understanding EMS response times. These papers establish that EMS response times represent a welfare-relevant outcome with large economic consequences. Wilde (2013) provides causal evidence, showing that, controlling for the endogeneity of dispatching via instrumental variables, each additional minute of EMS response time raises 90-day mortality by roughly 1 percent, an effect not evident in the OLS results.

Brent and Beland (2020) and Courtemanche et al. (2019) identify two mechanisms slowing response times. Traffic congestion imposes an unpriced externality on EMS worth an estimated \$95–\$285 million annually in California alone, while the ACA’s insurance expansion generated a demand shock that slowed ambulance response times by roughly 24 percent, costs that standard value-of-time and insurance welfare calculations, respectively, fail to capture.

This paper examines whether New York City’s congestion pricing policy - the nation’s first comprehensive cordon-based congestion pricing program and implemented on January 5, 2025 - improved EMS response times in the priced area. Using a difference-in-discontinuities design that exploits the sharp geographic boundary of the congestion relief zone at 60th Street, we find that congestion pricing generated substantial, localized reductions in vehicular traffic. Passenger vehicle density fell by approximately 21 percent and truck traffic by 18 percent near the boundary, providing first-stage evidence that the policy affected mobility in the priced area. These traffic reductions translated into clear improvements in EMS performance. Travel time to incident scenes declined by approximately 7–9 seconds, hospital transport time fell by roughly 54–59 seconds, and total EMS travel time decreased by 63–70 seconds, a reduction of approximately 5–6 percent relative to pre-policy baselines.

The impacts emerged quickly following the January 2025 congestion price implementation, concentrated near the zone boundary, and show little evidence of displacement to adjacent untreated areas. An FDNY directive issued in March 2025 requiring ambulances to transport patients to the nearest hospital rather than a preferred facility, which substantially altered transport patterns

citywide and confounds standard difference-in-differences estimates, represents an important confounding factor. After accounting for this concurrent policy change, the evidence consistently points to important improvements in EMS response performance within the congestion zone, suggesting that congestion pricing delivers emergency response benefits that existing cost-benefit frameworks have not previously captured.

This paper contributes to two largely separate literatures that have not previously been connected. In the literature on EMS responses, the paper is the first to examine congestion pricing as an exogenous source of variation and a policy lever for improving ambulance performance, extending Brent and Beland (2020), who document traffic congestion as an unpriced externality that slows emergency response. Brent and Beland (2020) establish the problem and estimate its aggregate cost. This paper identifies a solution, providing quasi-experimental evidence on whether one of the most effective congestion-reduction tools actually delivers benefits to emergency responders.

In the congestion pricing literature, the paper extends a well-developed body of work on the direct effects of congestion pricing on traffic, travel speeds, accidents, and air quality, reviewed extensively in the context of London, Stockholm, Milan, and Singapore, by identifying a novel category of external benefit that prior evaluations have not quantified: faster emergency medical response. This is important because cost-benefit analyses of congestion pricing policies, which typically weigh implementation and enforcement costs against commuter time savings and environmental gains, will systematically understate net social benefits if EMS response-time improvements are omitted. The paper also makes a methodological contribution by demonstrating the importance of accounting for concurrent policy changes, specifically the FDNY nearest-hospital directive, that can confound identification of congestion pricing effects, a practical lesson for future evaluations of urban transportation policies in complex institutional environments.

Literature Review

EMS Response Times, Health Outcomes, and Economic Costs

A growing body of economic research establishes that emergency medical service (EMS) response times constitute an important outcome in their own right, an outcome that sits at the intersection of public good provision, externality measurement, health insurance market design, and capacity constraints. Three interrelated contributions, Wilde (2013), Brent and Beland (2020), and Courtemanche et al. (2019), motivate an economic analysis of EMS response times.

Identification represents the primary challenge in analyzing the economic importance of EMS response times. Conventional OLS estimates of the effect of response time on patient mortality consistently find no statistically significant relationship (Pons et al., 2005; Newgard et al., 2010). Wilde (2013) demonstrates that this null result reflects endogeneity rather than an absence of effect: dispatchers allocate resources strategically, prioritizing the most severe cases and inducing a negative correlation between response time and unobserved incident severity. Because the researcher cannot fully observe the information available to the dispatcher, which may include something as

subtle as the caller’s tone of voice, OLS estimates of the effect of response time on mortality are biased toward zero.

To address this endogeneity, Wilde (2013) proposes an instrumental variable strategy using the straight-line distance from the incident location to the nearest authorized EMS agency headquarters as an instrument for response time. Using detailed call-level data from Utah’s Bureau of Emergency Medical Services linked to mortality records, she finds that the IV estimates of the impact of a one-minute increase in response time on 90-day mortality is approximately 1.0 percentage point, an 8 to 17 percent increase relative to baseline mortality depending on the time horizon considered. The corresponding OLS estimate is statistically indistinguishable from zero. These findings are robust to conditioning on detailed illness and injury categories, census block group fixed effects, and provider fixed effects, and hold for a broad population of EMS patients rather than only cardiac arrest victims, who constitute only about 1 percent of all EMS calls but dominate the prior medical literature.

Beyond mortality, Wilde documents that longer response times significantly raise the probability of emergency department admission and are associated with higher assessed severity on ER arrival, consistent with a mechanism in which delayed arrival allows ongoing physiological deterioration that cannot be fully reversed by subsequent treatment. Response times do not appear to affect the number of procedures or total hospital expenditures conditional on admission, suggesting that the welfare cost of slower response times operate primarily through mortality and severity channels rather than through downstream resource use.

The welfare cost of traffic congestion is conventionally measured through the value of time lost by private drivers and, increasingly, through external effects on pollution, health, and house prices (Currie and Walker, 2011; Knittel et al., 2016; Ossokina and Verweij, 2015). Brent and Beland (2020) document a previously unrecognized channel: traffic congestion slows fire trucks and EMS vehicles responding to emergencies, generating an additional external cost that standard value-of-time frameworks systematically ignore.

Using incident-level data from over 1.3 million California fire department calls between 2008 and 2015 matched to high-resolution traffic data from the California Department of Transportation, a sample containing approximately 21 billion observations at five-minute intervals from 25,000 monitoring stations, Brent and Beland estimate the causal effect of pre-alarm traffic conditions on response times. Their identifying assumption is that lagged deviations from free-flow traffic in a given zip code, month, day of week, and hour of day are uncorrelated with unobservables affecting response times conditional on fixed effects. Because traffic is measured before the alarm occurs, reverse causality from the incident itself is ruled out by construction.

The estimated effects contain non-linearities. Traffic in the 9th and 10th deciles of the delay distribution significantly increases response times by approximately 0.135 and 0.191 minutes, respectively, relative to minimal traffic. Lower deciles have coefficients indistinguishable from zero. This right-tail concentration has important policy implications: average congestion reduction may do little to improve emergency response, whereas targeted interventions that reduce the incidence of

extreme congestion episodes are likely to be most effective. Results are consistent across metropolitan areas, call types (fires and EMS), and a number of robustness checks including metro-by-date fixed effects, alternative traffic variable definitions, and different temporal assignment windows.

Brent and Beland also explore effect heterogeneity along several dimensions. First, they find suggestive evidence of adaptation. Fire departments facing chronically high average traffic conditions exhibit smaller marginal response-time increases per unit of congestion, consistent with investment in additional stations or routes in response. Second, decomposing traffic into expected and unexpected components using moving average forecasts, they find that unexpected traffic drives the response-time effect while expected traffic does not, reinforcing the adaptation interpretation and suggesting that the social cost of unpredictable congestion episodes exceeds that of predictable ones. Third, the marginal effect of traffic is significantly smaller in zip codes with high-occupancy vehicle (HOV) lanes providing evidence that emergency vehicles benefit from access to less-congested lanes while other policies such as toll roads and rail transit do not produce statistically significant mitigation effects.

The paper further links response-time delays to external economic costs. Extreme traffic conditions increase fire property damages, particularly in non-urban areas where baseline response times are longer and congestion episodes are harder to anticipate. Combining estimates of the traffic-response time relationship with Wilde’s mortality elasticity, Brent and Beland calculate aggregate welfare losses from traffic-related EMS slowdowns of approximately \$95–\$285 million per year in California. This figure is several orders of magnitude smaller than the value of time lost to commuters, but it is still important for two reasons. First, this likely represents a substantial underestimate because it omits police and private ambulance responses, adaptive investment costs, and counties without traffic monitoring infrastructure. Second, it represents a cost that standard value-of-time calculations which rely on willingness-to-pay elicited from private drivers miss entirely, since emergency vehicles do not pay tolls and presumably have a far higher social value of time than ordinary commuters.

While Wilde (2013) and Brent and Beland (2020) identify supply-side and externality channels that slow response times, Courtemanche et al. (2019) document how policy-induced expansions in EMS demand can strain emergency response capacity. Exploiting temporal and geographic variation in the implementation of the Affordable Care Act (ACA) combined with pre-treatment county-level uninsured rates in a triple-difference identification framework, they estimate that full ACA implementation slowed ambulance response times by approximately 2.4 minutes, roughly 24 percent relative to the pre-reform mean. The effect is driven by an 8 to 9 percent increase in EMS dispatches as newly insured individuals who would not otherwise have called an ambulance contacted the system. The supply-side response was muted.

Data on EMS employees and salaries show no significant adjustment, consistent with the capital-intensive nature of ambulance services and the substantial training requirements for EMS personnel. Event-study estimates confirm the absence of pre-trends, and the slowdown was concentrated in the 8-, 13-, and 20-minute response-time thresholds rather than the 4-minute threshold most critical for

cardiac outcomes. Results imply a short-run elasticity of response time with respect to insurance enrollment of approximately 3.4, a size warranting explicit consideration in welfare evaluations of coverage expansions.

Taken together, these three papers establish that EMS response times are a significant and underappreciated component of economic welfare. Wilde (2013) provides the causal identification necessary to translate response-time changes into mortality and hospitalization outcomes. Brent and Beland (2020) document that traffic congestion imposes real costs through this channel that are invisible to standard externality accounting. Courtemanche et al. (2019) show that policies increasing insurance coverage, whose benefits are well-documented, also carry a supply-side cost through emergency response congestion, at least in the short run.

Together, these contributions imply that any comprehensive cost-benefit analysis of transportation policies, urban land use, or health insurance design should account for response-time effects as a distinct welfare channel. They also suggest a complementarity between public goods that economists have only begun to model formally (Albouy et al., 2020). The value of the emergency response system depends critically on whether roads are passable and whether system capacity is matched to insurance-induced demand.

Congestion Pricing and Driver Behavior

Congestion pricing has generally been shown to reduce traffic volumes entering priced urban zones. This reduction is most pronounced in the period immediately following implementation. Leape (2006) finds that the London Congestion Charge produced an immediate reduction in traffic entering central London after its introduction in early 2003. He describes the institutional design change of the London system, including the charging zone, enforcement technology, exemptions, revenue use, which was necessary for success. Croci (2016) compares London, Stockholm, and Milan and finds that each city experienced reductions in traffic or congestion after adopting urban road-pricing schemes. The magnitude and persistence of these effects varied across local contexts. London focused heavily on congestion targets while Stockholm combined congestion and environmental targets. Milan initially emphasized emissions management and later included broader congestion reduction goals.

Eliasson (2009) finds that Stockholm’s congestion charging system reduced traffic volumes and generated net welfare gains through overall time savings. The study conducts a formal cost-benefit analysis, weighing travel-time improvements, environmental benefits and implementation costs. Börjesson et al. (2012) show that traffic reductions persisted several years after implementation and that public acceptance increased after local resident experiences. Moreover, analyses of Stockholm’s system through the first five years post-adoption showed that behavioral responses and political support evolved after the initial trial period. These studies suggest that congestion pricing can reduce vehicle entries into priced zones, particularly in the short run. Further, they show that the magnitude and durability of the effect depends on enforcement, exemptions, pricing levels, and exactly how cities reallocate road space after implementation.

Metz (2018) examines congestion pricing systems in the urban areas of London, Stockholm, and Singapore and finds that congestion pricing significantly reduced traffic entering charged zones, generally by roughly 15 to 25 percent. Further this program improved travel speeds and transportation efficiency. However, the study also found that the positive impacts of reduced congestion can attract additional drivers over time and partially offset initial improvements. The study concludes that congestion pricing is only a partial urban congestion tool given that lower vehicle volumes do not always correlate with permanent reductions in measured congestion.

Duranton and Turner (2011) use U.S. metropolitan-area data on road capacity and vehicle travel to test whether building more roads reduces congestion. Their main finding is that increases in roadway capacity lead to roughly proportional increases in vehicle miles traveled. This leaves the issues of long-run congestion unsolved since expanding roads seems to induce additional driving through more trips by local residents, increased commercial traffic, and migration trends.

Recent work provides the first causal evidence on the effects of New York City’s congestion pricing program on traffic conditions. Using a generalized synthetic control approach and high-frequency Google Maps traffic data spanning through June 2025 (first six months post-adoption), Cook et al. (2025) find that the introduction of congestion pricing led to a substantial increase in traffic speeds within the central business district, on the order of 10–15 percent, with larger gains during peak congestion periods. These improvements also extended beyond the priced zone, with positive spillover effects on roads commonly used by drivers traveling to and from the central business district, leading to faster trips throughout the broader metropolitan area. Importantly, their findings highlight that congestion pricing can generate network-wide efficiency gains, rather than simply redistributing traffic across space. This evidence provides a useful benchmark for interpreting the first-stage effects in our analysis and supports the view that congestion pricing can deliver meaningful improvements in traffic conditions, particularly in the short run.

Congestion and EMS Response Times

Relatively little research examines how congestion affects EMS response times specifically in New York City (or other urban areas) given its’ diverse transportation mechanisms. Two common measures that have been investigated within the literature are: (1) dispatch location to scene time as well as (2) scene to hospital times. In regards to the former: Brent and Beland (2020) matched traffic congestion data to California fire department incident records to estimate how congestion affects emergency response times, framing slower responses as a negative externality (of traffic congestion). They utilize traffic conditions immediately preceding incidents. The results indicate that higher congestion is associated with longer response times.

Griffin and McGwin Jr (2013) employ a survey of EMS providers about their experiences with traffic congestion. Survey respondents reported that congestion added approximately 10 minutes of response time on average. Tanaka et al. (2013) utilized a longitudinal study to analyze traffic patterns over an 8-year period (April 1, 2003 through March 31, 2011) in Kanazawa City (capital of Ishikawa Prefecture in central Japan). The authors evaluate Japan’s FAST traffic-signal pre-

emption system. The FAST (Fast Emergency Vehicle Preemption System) traffic-signal priority system is designed to reduce emergency vehicle travel time by changing signal timing in favor of an approaching ambulance. By evaluating FAST-equipped versus non-FAST-equipped ambulance dispatches for out-of-hospital cardiac arrest, their results found that FAST-equipped dispatches had shorter median response times. FAST-equipped dispatches were also associated with higher 1-year survival rates, including in analyses that restricted to a subgroup of cases.

Research that examines transport from scene to hospital times include Fukushima and Moriya (2021) who use ambulance GPS (Global Positioning System) data and EMS (Emergency Medical Services) records to evaluate the time interval from fire department departure to hospital arrival. They focus on quantifying potential time savings during ambulance running, rather than isolating a causal effect of congestion by comparing a preferred or shortest feasible interval for each trip. To do this they take GPS probe data from each ambulance and match it to the EMS record for 68 transports.

For cases that were not dispatched from the nearest fire department, they compute a simulated time as if the ambulance had been dispatched from the nearest fire department (which could be due to traffic flows or EMS coverage and availability constraints), then compare those simulated times to the actual non-nearest dispatch times. Further, they identify ambulance runs that passed a speed reduction zone (and more generally extract congested road segments), then simulate a scenario where those constraints are avoided producing a shorter simulated running time. They treat the simulated scenario (nearest dispatch, and/or improved congestion or avoiding speed reduction zones) as a benchmark for time savings and then test whether the observed vs simulated differences are statistically significant. They do not utilize any sort of randomized or quasi-experimental congestion effect.

Gorgens et al. (2024) tests whether traffic patterns (peak vs off-peak) affect the accuracy of EMS prenotified estimated transport interval (ETI) for trauma activations. They use data from the Northwell Health Level I urban trauma center in Manhasset, New York. They describe peak hours as trauma transports where EMS arrived at the hospital between 06:00 to 10:00 and 16:00 to 20:00, Monday through Friday. Their results show that EMS transport time estimates underestimated the actual arrival time (too optimistic).

The authors did not find evidence that peak traffic hours made estimates systematically less accurate than off-peak hours. EMS underestimated ETI (estimated transport interval) 55% overall and similar to off-peak hours. In a similar vein Neeki et al. (2016) compared EMS estimated time of arrival (ETA) to actual time of arrival (TOA) for trauma transports to a Level II trauma center in Colton, California (Arrowhead Regional Medical Center) and quantify underestimation. In their sample, the median ETA was 10 minutes versus a median actual TOA of 22 minutes (median underestimation about 9 minutes), and the discrepancy varied by month and time of day.

Our study contributes to two related but largely separate literatures. First, in the congestion pricing literature, it provides the first empirical evidence on how New York City’s congestion pricing program affects emergency medical service response performance, thereby identifying a novel

and policy-relevant external benefit of congestion pricing. While prior work has focused primarily on traffic volumes, travel speeds, and environmental outcomes, our analysis highlights an additional channel through which congestion pricing can improve welfare—by enhancing the speed and reliability of emergency response. Second, in the literature on EMS response times, we identify congestion pricing as a new source of exogenous variation in traffic conditions and show how reductions in congestion translate into measurable improvements in ambulance travel times. Methodologically, we implement a difference-in-discontinuities design that exploits the sharp geographic boundary of the congestion relief zone, enabling credible causal identification while accounting for pre-existing spatial discontinuities and concurrent system-wide changes. By integrating high-frequency traffic data with incident-level EMS records and examining both dynamic and spatial patterns of adjustment, this study provides a comprehensive view of how congestion pricing in general, and NYC’s first-in-the-nation program in particular, affects mobility and public service delivery.

Empirical Analysis

Data

We use EMS Incident Dispatch Data provided by the Fire Department of New York City through the NYC Open Data portal for our main analyses of EMS response times. These data are generated by the city’s computer-aided dispatch (CAD) system and contain detailed records of emergency medical service (EMS) responses to 911 calls that result in ambulance dispatches. The data capture the full time line of each EMS response, from call creation to completion, and includes information on dispatch decisions, unit assignments, and response times.

To comply with the Health Insurance Portability and Accountability Act (HIPAA), the data do not include exact geographic coordinates of health incidents or individual identifiers. Instead, incidents are geo-referenced using five different administrative areas, including the ZIP code, police precinct, community district, school district, and congressional district where each incident requiring an EMS call occurred. This means we need to estimate the approximate location of each incident. We combine information from all five administrative areas to identify more spatially refined “intersection polygons” where each incident occurred. We construct these intersection polygons by overlaying the map area defined by each administrative boundary for each EMS incident and identifying areas that appear in all five areas.

Table 1 contains information about each of these administrative areas in Manhattan included in the EMS data, including the number of individual units, the average size of each type of unit, and the average number of EMS calls in each unit over the sample period for each. Each administrative unit identified in the data file represents a potential spatial treatment area for the empirical analysis. These administrative units are listed on Table 1 by size from largest to smallest. Even using ZIP Code areas as the treatment area leads to a relatively imprecise approximation of the location of each incident covering more than 1 square kilometer and containing more than 10,000 EMS calls.

Table 1: Manhattan Spatial Administrative Units

Spatial Unit	Number	Mean Area (km ²)	Mean Dispatches
Congressional District	3	18.4	105,614
School District	6	9.2	75,439
City Council District	10	5.5	48,007
Community District	12	4.3	35,466
Police Precinct	23	2.4	22,960
ZIP Code	44	1.2	10,561
Intersection Polygon	188	0.3	2,708

Identifying areas where these five administrative layers overlap allows us to determine the location of each incident far more accurately than using any of the individual tagged areas alone. The row labeled *Intersection Polygon* on Table 1 summarizes the actual spatial treatment area used in the empirical analysis. Our 188 intersection polygons are substantially smaller than ZIP Code areas, the next largest potential treatment area, and contain many fewer EMS calls. Using intersection polygons significantly sharpens our estimate of the location of each incident.¹

Figure 1 shows the location of each intersection polygon and the CRZ boundary as a dashed red line. In total, overlaying the areas of each administrative unit yields 188 distinct intersection polygons containing all incidents in our sample. These intersection polygons completely partition all of Manhattan into discrete areas. Figure 1 also shows the number of EMS calls in each intersection polygon, with darker shading representing fewer calls.

The intersection polygon for each incident allows us to accurately compute the approximate distance from each incident to the CRZ boundary, which is central to our difference-in-discontinuities identification strategy. Specifically, for each incident we compute the driving distance to the CRZ boundary from: (a) the geometric centroid of the intersection polygon; (b) a population-weighted centroid of the polygon; (c) a uniform random sample of points within the polygon; and (d) population-weighted random sample of points within each polygon. While the absence of exact coordinates introduces some measurement error in this distance calculation, this error is likely to be classical and should not bias the estimated discontinuity effects. We show below that our estimates are robust across these alternative distance estimates.

To document first-stage changes in traffic and mobility conditions after the CRZ implementation, we supplement the EMS data with traffic density measures from publicly available New York City traffic cameras deployed at fixed locations across Manhattan. C2Smart, a transportation research center at New York University, has developed a machine learning algorithm that processes these camera snapshots to classify observed objects as passenger vehicles, trucks, pedestrians, or

¹Powell (2025) analyzes the impact of congestion pricing on EMT response times to incidents in Manhattan using a difference-in-differences model and an event study comparing call times identified by Community Districts before and after the introduction of congestion pricing inside and outside the CRZ. Powell (2025) reports a 15 second reduction in EMS times to incidents located inside the CRZ after implementation of congestion pricing. We use a sharper identification based on RD methods and a much smaller spatial treatment area and investigate first-stage effects. We also focus on total response times and time from incident to hospital.

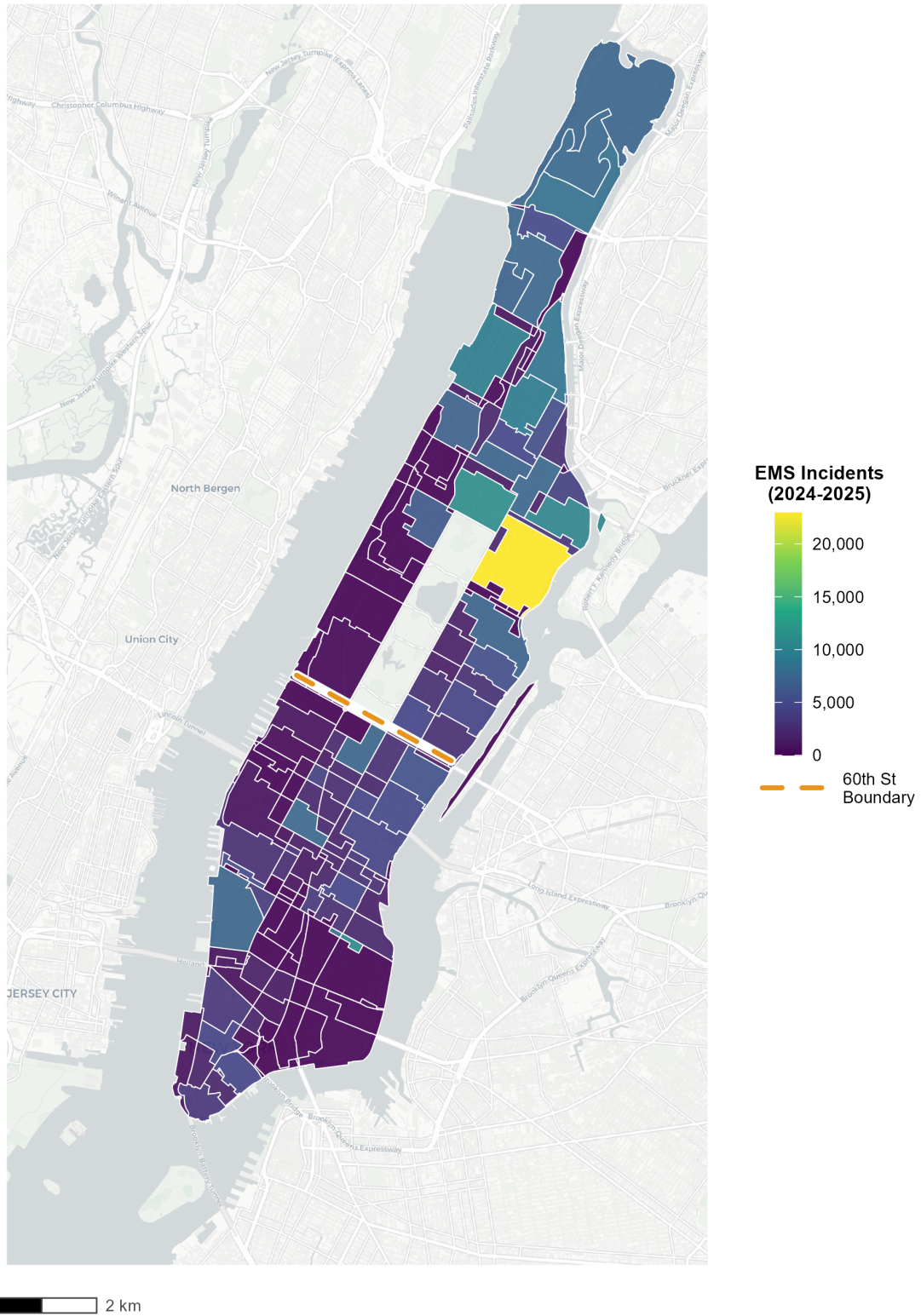


Figure 1: Intersection Polygons and EMS Incidents

bicycles, reporting the average number of each road-user type visible per snapshot (Zuo et al., 2025).²

Each camera records a snapshot observation every 15 minutes, rather than continuous monitoring over the time period. We use these camera snapshot data to estimate how congestion pricing altered traffic density and modal composition at the 60th Street boundary under the same difference-in-discontinuities framework applied to EMS outcomes. These first-stage estimates provide evidence consistent with the mechanisms through which congestion pricing may affect EMS response times and help frame the magnitude of the downstream effects documented in the following section.

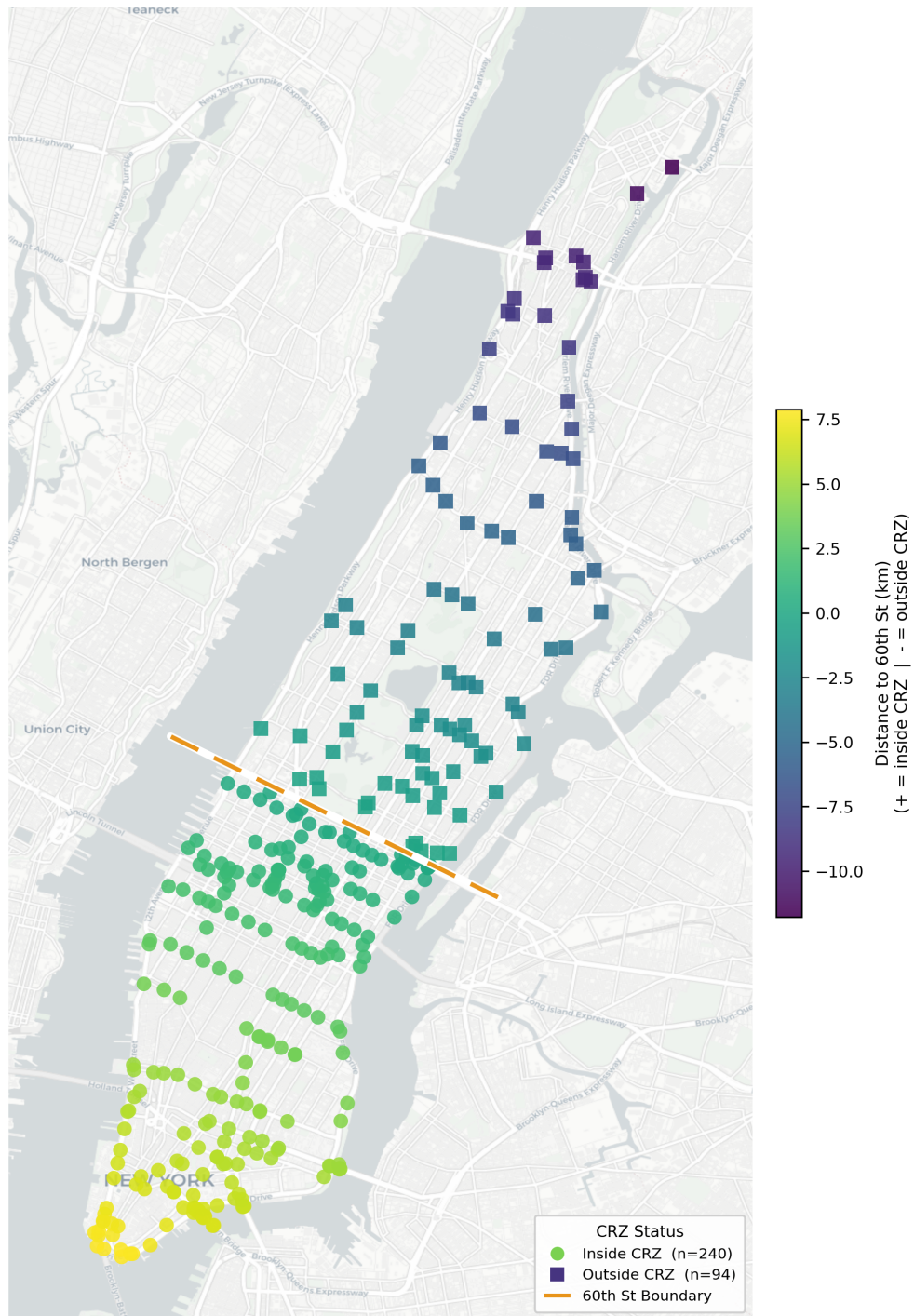
Taken together, the EMS and traffic data provide high-frequency, spatially granular measures of both response outcomes and underlying traffic conditions, enabling a detailed analysis of congestion pricing using the difference-in-discontinuities framework. Figure 2 contains a map of the congestion relief zone boundary and the locations of camera sensors used in this analysis. The colors show distance from each sensor to the 60th Street congestion zone cutoff. From Figure 2, a large number of sensors are located just above and below 60th Street, facilitating our difference-in-discontinuities approach.

The primary analyses utilize data from the full calendar years 2024–2025, which contains the implementation of congestion pricing in New York City on January 5, 2025. We further limit the sample to incidents occurring in Manhattan, which contains both the congestion relief zone and the immediately adjacent control area north of 60th Street, and restrict the sample to observations with valid time measurements.

A typical EMS response cycle consists of several stages. When a 911 call reporting a medical emergency is placed, the operator asks a series of basic questions about the patient’s condition and exact location. Then the call is transferred to EMS dispatch. A medical dispatcher uses a specific protocol to classify the emergency and assign it a priority level. For example, cardiac arrest calls receive the highest priority, and minor injuries are assigned lower priority levels. Depending on the severity of the incident, the dispatcher sends the closest appropriate unit and determines whether the response requires lights and sirens. The ambulance then travels from its post or prior assignment to the incident location. On the scene paramedics assess the severity of the incident, stabilize the patient, determine whether transportation is necessary, and identify the most appropriate hospital.

For example, major trauma patients are typically transported to Level I trauma centers, and stroke patients are taken to certified stroke centers. The patient is then transported from the incident site to the hospital, where EMS transfer them to the hospital staff. Finally, the ambulance crew cleans equipment and restocks supplies, completes paperwork, and then becomes available for the next call.

²Although camera locations are fixed, the cameras themselves are pan-tilt-zoom (PTZ) units whose field of view can shift over time. C2Smart’s processing pipeline addresses this by clustering frames from each camera by viewpoint similarity and retaining only frames from the camera’s dominant (most frequently observed) viewpoint, discarding the rest. As a result, some periods have no usable observations if the camera was not pointed at its dominant view during that interval. See Zuo et al. (2025) for details on this viewpoint-normalization procedure.



Source: C2Smart / NYC DOT sensor network. Boundary at 60th Street, Manhattan.
(c) OpenStreetMap contributors / CartoDB

Figure 2: C2Smart Traffic Sensor Locations

Appendix Figure A1 shows the distribution of EMS stations across Manhattan and Appendix Figure A2 shows the distribution of hospitals. EMS stations are relatively uniformly distributed across Manhattan. Hospitals tend to cluster on the east side of the island with a particularly large cluster east of Central Park.

We focus on three key EMS response durations: travel time to incident, travel time from incident to hospital, and total travel time. Travel time to incident is defined as the time elapsed in seconds between the moment an EMS unit is assigned to the case and the moment it signals that it has arrived at the incident location. Travel time to hospital measures the time between leaving the scene and arriving at the hospital destination. Travel time to hospital is not directly recorded but is constructed using timestamps for (i) departure from the scene and (ii) arrival at the receiving hospital. Total travel time is defined as the sum of travel time to incident and travel time to hospital, and is available only for incidents that result in hospital transport.

The spatial precision provided by the availability of the exact latitude and longitude coordinates for each traffic camera data allows us to construct a measure of the network distance from each camera sensor to the 60th Street CRZ boundary for our difference-in-discontinuities design. We compute this network distance through the OSMnx Python library (Boeing, 2017), which constructs a network representation of the street and pedestrian network from OpenStreetMap data and calculates the shortest-path distance along the actual road or walking network between each observation unit and the boundary.

Because the OSMnx library supports separate network representations for driving, walking, and cycling, the specific network type used to compute distance is matched to the outcome variable: distance along the driving network is used for vehicle and truck outcomes, distance along the walking network is used for pedestrian outcomes, and distance along the cycling network is used for bicycle outcomes.

Table 2 contains summary statistics for the EMS and camera data. Mean travel time to the incident is approximately 475 seconds, just under 8 minutes, inside the congestion relief zone and 503 seconds outside. Mean hospital transport time is approximately 738 seconds, just over 12 minutes, inside the zone and 672 seconds outside. It takes EMS more time to transport a patient to a hospital than to arrive on the incident scene.

Several factors likely explain this disparity. First, EMS dispatchers usually pick the geographically closest available ambulance to scene. Moreover, EMS units themselves are not stationary but strategically deployed around the city in order to minimize the response time. Hospitals, however, are stationary, and after stabilizing a patient the ambulance needs to transport them to the nearest appropriate hospital (for example, cardiac patients to specialized cardiac facilities), which might not be the geographically closest one.

Second, ambulances traveling to an accident scene might drive more aggressively and are more likely to use light and sirens because in some cases it is crucial to get to the patient as fast as possible; moreover, response time to the scene is one of the most important performance metrics which might provide additional incentives for the EMS crew. Transportation to the hospital may

Table 2: Summary Statistics

	Inside CRZ	Outside CRZ
Panel A: EMS Response		
<i>Time to Incident (sec)</i>		
Mean	474.90 (277.05)	503.12 (299.66)
<i>Time to Hospital (sec)</i>		
Mean	737.81 (345.21)	671.71 (319.62)
<i>Total Travel Time (sec)</i>		
Mean	1205.74 (479.75)	1168.71 (463.85)
Observations	260,706	263,013
Panel B: C2Smart Traffic Cameras		
<i>Passenger Vehicle Density (15-min periods)</i>		
Mean	3.59 (4.87)	4.34 (5.82)
<i>Truck Density (15-min periods)</i>		
Mean	0.44 (0.85)	0.41 (0.80)
<i>Pedestrian Density (15-min periods)</i>		
Mean	3.33 (4.54)	2.45 (3.53)
<i>Bike Density (15-min periods)</i>		
Mean	0.01 (0.08)	0.00 (0.07)
Observations	871,357	348,679

Notes: Standard deviations in parentheses. EMS times in seconds; C2Smart density metrics reflect the average number of road users of each type visible per 15-minute snapshot interval (Zuo et al., 2025).

involve more cautious driving, especially when the patient is already stabilized or when medical care is being administered during transport.

From Panel B of Table 2, the C2Smart camera data covers 871,357 fifteen-minute intervals periods inside the CRZ and 348,679 outside it, roughly 2.5 times as many observations inside the CRZ, reflecting denser camera coverage there, visible on Figure 2. The most notable pattern appears in the pedestrian density, where inside-CRZ locations average 3.33 per 15-minute interval compared to 2.45 outside, about 36% higher foot traffic inside the zone. Passenger vehicle density runs in the opposite direction: 3.59 inside versus 4.34 outside, suggesting somewhat lower car traffic within the CRZ, which is consistent with what a congestion or restricted zone policy would be expected to produce.

Truck density is nearly identical on both sides (0.44 vs 0.41), suggesting the zone boundary has little differential effect on commercial vehicle presence, at least as captured by these cameras. Bike density is negligible in both areas, essentially zero (0.01 inside, 0.00 outside), with very low standard deviations, suggesting cycling is rare enough in this setting that the camera data cannot meaningfully capture it. In all categories the standard deviations are large relative to the means, indicating considerable variation across camera locations and time periods, which is typical of traffic density data.

Empirical Strategy

To estimate the causal effect of New York City’s congestion pricing policy on EMS response times, we adopt a difference-in-discontinuities (DD-RD) design (Tramontin Shinoki et al., 2024) that exploits the sharp geographic boundary of the congestion relief zone (CRZ) at 60th Street in Manhattan. As noted earlier, the policy, implemented on January 5, 2025, introduced a discrete and substantial increase in the cost of vehicle entry for locations south of (and including) 60th Street, while locations immediately north remained untreated. This creates a sharp spatial discontinuity in exposure to congestion pricing. Our approach compares EMS outcomes just south of the boundary (treated) to those just north (control), before and after the policy. This framework combines the strengths of regression discontinuity (RD) and difference-in-differences (DD), allowing us to difference out any pre-existing discontinuities at the boundary.

Relying on localized comparisons around a clearly defined and exogenous policy boundary reduces concerns about confounding differences across neighborhoods. Note that the approach does not require distant control areas or strong assumptions about trends across locations. The DD-RD design is particularly well-suited here because areas immediately on either side of the 60th Street boundary are highly comparable in terms of infrastructure, hospital access, and EMS deployment, yet differ discretely in their exposure to the congestion pricing policy.

Our baseline DD-RD specification takes the following flexible form:

$$\begin{aligned}
Y_{ist} = & \alpha + \beta [\mathbf{1}(d_i \geq 0) \cdot \text{Post}_t] + [f_1(d_i) \cdot \mathbf{1}(d_i \geq 0)] \\
& + [f_2(d_i) \cdot \mathbf{1}(d_i < 0)] + [f_3(d_i) \cdot \mathbf{1}(d_i \geq 0) \cdot \text{Post}_t] \\
& + [f_4(d_i) \cdot \mathbf{1}(d_i < 0) \cdot \text{Post}_t] + Z_{ist} + \lambda_t + \mu_s + \varepsilon_{ist}
\end{aligned} \tag{1}$$

where Y_{ist} denotes our outcomes of interest, measuring EMS response times to the scene of the incident, from incident to the hospital, and total travel time, for incident i in dispatch area s at time t . The running variable is the signed distance to the 60th Street boundary, $d \geq 0$ for the CRZ, south of the boundary and $d < 0$ for the non-tolled area, north of the boundary, and Post indicates the post-January 5, 2025 period. The functions $f_k(\cdot)$ allow for flexible spatial trends on either side of the boundary. In our baseline specification, we model these as linear functions with slopes allowed to differ across sides of the boundary and between pre- and post-policy periods, and we assess robustness to alternative functional forms and nonparametric specifications.

The vector λ_t denotes time (year-by-month, day-of-week) fixed effects capturing common shocks such as seasonality and system-wide operational changes such as the March 2025 directive requiring NYC ambulances to transport patients to closest hospitals as determined by the dispatch system. The μ_s reflect spatial fixed effects which absorb persistent differences in EMS infrastructure, hospital proximity, and baseline traffic conditions across dispatch zones while preserving within-boundary variation for identification.³

Additionally, we control for incident-specific observable characteristics through fixed effects for call type (detailed categories assigned by 911 dispatchers to classify the nature of the medical emergency), severity (8-tier severity code ranging from a top-priority life-threatening emergency to a minor non-life-threatening incident), and disposition of the emergency response (e.g. hospital transport, treated at scene, patient refusal, etc.).⁴

Estimates are not sensitive to these controls, consistent with the identifying assumption that incident characteristics vary smoothly across the boundary and that congestion pricing does not induce sorting or compositional changes in EMS calls near the cutoff. Standard errors are clustered at the level of the polygon area to account for within-location correlation over time.⁵

To assign EMS incidents to treated or control exposure, we account not only for the location of the incident but also for the spatial distribution of travel along the ambulance route. Because EMS response can involve travel across both sides of the congestion zone boundary, classifying incidents solely based on incident location may misrepresent actual exposure to congestion conditions. Instead, for each outcome we assign treatment based on the portion of the relevant travel route that

³Dispatch areas are geographic subdivisions that function as localized deployment territories used by the Fire Department of New York (FDNY) to manage and route EMS operations across the city. These fixed effects absorb any time-invariant differences across dispatch zones related to, for instance, workload, crew composition, local routing patterns and such.

⁴While equivalent to estimating separate regression discontinuity models before and after the policy and differencing the estimated discontinuities, estimating the DD-RD effect within a single equation facilitates straightforward statistical inference.

⁵Note that the intersecting polygon is the smallest geographic area that can be imputed for an EMS incident, and which we use to compute the distance to the boundary, thus generating the variation in the running variable. Inference is not sensitive to alternately clustering at the level of the dispatch area or the zip code.

lies within the congestion zone. Specifically, for response time to the incident, we construct the shortest-path route between the incident location and the nearest EMS or fire station, and classify the observation as treated if the majority of this route lies within the congestion zone.

For hospital transport time, we similarly construct the shortest-path route from the incident location to the nearest hospital and assign treatment based on where the majority of that route occurs. For total response time, we combine both segments—the route from station to incident and from incident to hospital—and assign treatment based on which side of the boundary accounts for the greater share of the combined route. This route-based classification better captures the extent to which EMS travel is exposed to congestion pricing and associated traffic conditions.⁶

The coefficient of interest, β , captures the change in the discontinuity at the boundary in the post-policy period relative to the pre-policy period. Identification relies on the assumption that, absent the congestion pricing policy, EMS response times would have evolved smoothly across the boundary. By differencing out any baseline discontinuity, the DD–RD approach relaxes the standard RD requirement that outcomes be identical at the threshold in levels.⁷

In our setting, identification is grounded in two key features. First is that the policy boundary is arguably exogenous; the 60th Street cutoff is administratively defined and unrelated to EMS operations or localized shocks to emergency demand. Second, restricting attention to observations near the boundary ensures that treated and control locations are similar in both observed and unobserved characteristics. To assess the plausibility of these assumptions, we conduct standard tests including: (1) placebo checks based on “pseudo-treatment” effects and falsification outcomes that should not be impacted by the congestion pricing program (CPP); (2) covariate balancing across the boundary pre- and post-CPP; and (3) sensitivity to bandwidth and alternate functional forms.

We extend the baseline model specified in equation (1) in various ways to address specific issues. First, we implement the same DD–RD framework to study the effects of congestion pricing on truck and passenger vehicle traffic, as well as bicycle traffic and pedestrian activity. Documenting these effects is critical for interpreting our main estimates, as any impact of congestion pricing on EMS response times must operate through changes in traffic conditions, congestion, and broader mobility patterns.

Establishing that the policy generates sharp and localized changes in these underlying mechanisms at the 60th Street boundary provides a key validation of the research design and strengthens the credibility of the causal interpretation of downstream effects on EMS performance. Moreover, the magnitude and spatial pattern of these first-stage effects offer important context for understanding the size and heterogeneity of estimated impacts on response and transport times.

Second, because the discontinuity is spatial rather than temporal, we can estimate event-time

⁶With respect to travel to the incident (hospital), approximately 2.9 (3.5) percent of incidents within the CRZ involved a route which traversed the boundary and mostly took place outside the CRZ; these would be assigned to the control. And, approximately 0.2 (0.3) percent of incidents north of the CRZ involved a route which traversed the boundary and mostly took place inside the CRZ; these are more appropriately assigned to the treatment.

⁷Since Central Park spans north from 59th Street to 110th Street, persistent baseline discontinuity in mobility patterns at 60th Street are to be expected, which are netted out from the post-policy period in the DD–RD model.

DD-RD models that trace the dynamic evolution of treatment effects. Specifically, we allow the treatment effect to vary by lapsed time relative to policy implementation to distinguish immediate short-run effects (e.g., traffic adjustment) vs. medium-run responses (e.g., behavioral adaptation of drivers and EMS deployment).⁸

Third, we extend the baseline difference-in-discontinuities framework to examine spatial displacement and to more directly probe the mechanisms linking congestion pricing to EMS performance. While the primary design identifies localized changes in traffic conditions at the 60th Street boundary, congestion pricing may induce drivers to reroute to nearby or more distant non-tolled areas, shifting congestion rather than reducing it overall. To evaluate these dynamics, we estimate a “spatial event study”, allowing treatment effects to vary flexibly across distance bands on both sides as follows:

$$Y_{ist} = \alpha + \sum_{k \neq k_0} \beta_k [\mathbf{1}(d_i \in D_k) \cdot \text{Post}_t] + [f_1(d_i) \cdot \mathbf{1}(d_i \geq 0)] + [f_2(d_i) \cdot \mathbf{1}(d_i < 0)] + Z_{ist} + \lambda_t + \mu_s + \varepsilon_{ist} \quad (2)$$

where D_k denotes mutually exclusive distance intervals defined over the support of the signed distance variable, and one interval k_0 , the band immediately outside the congestion zone boundary, is omitted as the reference category. Each coefficient β_k captures the change in EMS outcomes in distance band k in the post-policy period relative to the corresponding change in the reference band.

This approach serves two complementary purposes. First, it provides evidence on mechanisms by tracing how first-stage changes in traffic patterns propagate across space and whether these spatial patterns align with changes in EMS outcomes. Second, it allows us to test for spatial reallocation of congestion by examining whether reductions within the congestion zone are offset by increases in adjacent or more distant areas. This distinction is critical for interpretation: improvements in response times within the zone may be attenuated if congestion increases along routes used by EMS vehicles outside the zone. By jointly examining spatial variation in mobility and EMS outcomes, we can distinguish between true improvements in traffic conditions and the spatial reallocation of congestion.

Finally, as a supplementary exercise, we estimate a difference-in-differences (DD) model that compares outcomes in the congestion relief zone to those in nearby non-treated areas over time. This model takes the form

$$Y_{ist} = \alpha + \delta [\mathbf{1}(\text{Treat}_s) \cdot \text{Post}_t] + Z_{ist} + \lambda_t + \mu_s + \varepsilon_{ist} \quad (3)$$

where Treat_s is an indicator equal to one if the incident lies within the congestion relief zone, and

⁸For the EMS data, we trace out dynamic effects by progressively adding each additional month of data over the post-policy window; that is, we estimate separate treatment effects for the first month, over the first two months, over the first three months, and so on, post-congestion pricing. For the C2Smart traffic camera data, which are available only for a period of nine weeks post-policy, we trace out these dynamics by progressively adding two-weeks of data over the post-policy window.

zero otherwise. All other variables are defined as above. The coefficient of interest, δ , captures the average change in EMS response outcomes in treated areas relative to control areas following the implementation of congestion pricing. While the DD–RD framework provides a highly credible local estimate of the policy effect at the boundary, the DD specification leverages broader spatial variation and allows us to complement the boundary-based analysis with a more aggregate comparison.

This approach serves two purposes. First, it provides a benchmark estimate based on a more conventional research design that relies on parallel trends rather than continuity at the boundary. Second, it offers descriptive evidence on how average outcomes evolve across a wider set of treated and untreated areas, including potential spatial heterogeneity and broader system-wide responses. However, because the DD design compares geographically distinct areas, it may also capture the influence of concurrent policy changes and differential trends across locations. Throughout, we interpret estimates from the DD–RD design as providing the primary causal estimates of the policy’s impact, while the DD and spatial analyses are used to provide complementary descriptive evidence on broader spatial and temporal patterns.

Results

First-Stage Effects on Traffic and Mobility

We begin by presenting first-stage evidence on how congestion pricing affected traffic volumes and broader mobility patterns near the 60th Street boundary in Table 3, based on the DD–RD model specified in Equation (1). Because the DD–RD design identifies local treatment effects at the policy boundary, the resulting estimates should be interpreted as reflecting changes in traffic conditions (and EMS performance) in areas near the 60th Street cutoff.

Table 3: C2Smart Traffic Cameras: Difference-in-Discontinuities Estimates				
	Passenger Vehicle	Truck	Pedestrian	Bike
DD–RD Estimate	-0.7846***	-0.0779***	0.4148***	0.0011***
Std. Error	(0.0224)	(0.0041)	(0.0192)	(0.0004)
Effect (%)	-20.6%	-18.1%	13.5%	19.8%
Pre-treatment Mean	3.8008	0.4308	3.0752	0.0054
Observations	2,360,613	2,360,613	2,360,613	2,360,613
<i>Notes:</i> DD–RD estimate (SE in parentheses), clustered on sensor. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.				

Table 3 presents first-stage evidence on how congestion pricing altered traffic density and mobility patterns near the 60th Street boundary. The results confirm that the policy generated sharp, statistically significant, and economically significant changes in the composition of street-level activity within the congestion relief zone. Passenger vehicle density fell by approximately 21 percent relative to the pre-treatment mean, and truck density declined by a similar margin of 18 percent, consistent with the toll inducing a broad reduction in motorized vehicle entries into the priced area.

These reductions are precisely estimated, with standard errors clustered at the sensor level yielding highly significant estimates for both vehicle categories.

In contrast, non-motorized activity increased substantially within the zone. Pedestrian density rose by roughly 14 percent, consistent with some commuters and visitors substituting walking for driving, while bicycle density increased by nearly 20 percent, suggesting large modal substitution toward walking and cycling. Taken together, these patterns paint a clear picture of the response to the toll. Drivers avoided the congestion zone, and at least a portion redirected their trips toward non-motorized alternatives. From the perspective of EMS operations, the vehicle traffic reductions are important, as they directly reduce the density of competing traffic that ambulances must navigate en route to incident scenes and hospitals.

The increases in pedestrian and bicycle activity, however, introduce a potentially offsetting influence, as greater street-level activity may increase intersection friction and slow ambulance movement at crossing points, a consideration that tempers but does not overturn the net positive implications of the first-stage results for emergency response performance.

Figure 3 shows the difference-in-discontinuities graphs, with observations on the plot representing residualized differences after conditioning out the fixed effects in the outcome of interest between the post-CPP and pre-CPP periods. These patterns are also consistent with substitution away from driving toward alternative modes of transportation in response to the toll, and they confirm that the policy generated substantial, localized changes in the density and composition of traffic at the CRZ boundary.

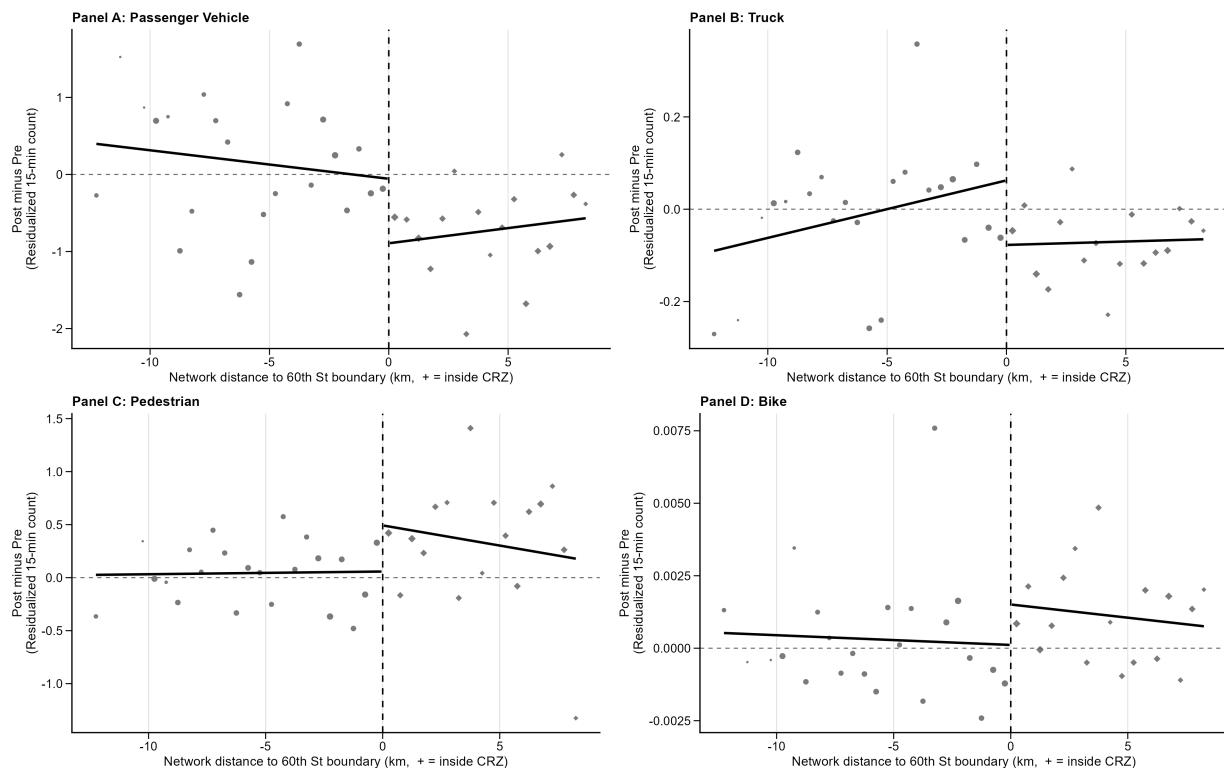


Figure 3: Difference-in-Discontinuities in Traffic and Mobility

Disaggregating the first-stage estimates by weekday and weekend periods, reported in Table 4, reveals a broadly similar pattern, with some informative heterogeneity across modes of transportation. Reductions in passenger and truck traffic are evident in both periods, with somewhat larger declines observed on weekends (26–27 percent vs. 15–18 percent). This pattern is consistent with greater price sensitivity in discretionary travel, as weekend trips are more likely to involve flexible, non-work-related activities that can be postponed, rerouted, or shifted to alternative modes.

In contrast, weekday travel is more heavily driven by commuting and commercial activity, which may be less responsive to pricing due to schedule constraints. Correspondingly, both pedestrian and bicycle activity increase within the congestion zone in each period, indicating substitution away from driving toward alternative modes of transportation. There are some nuanced differences, with pedestrian activity rising somewhat less strongly on weekends (8 percent on weekends vs. 16 percent on weekdays), while bicycle traffic exhibiting slightly larger increases during weekends (23 percent vs. 19 percent).

In summary, these results indicate that congestion pricing induces substantial and persistent changes in traffic and mobility throughout the week, while also highlighting modest differences in how individuals adjust travel behavior across contexts. From the perspective of EMS operations, these first-stage patterns are particularly informative: reductions in vehicular traffic—especially private cars and trucks—are likely to alleviate congestion-related delays faced by ambulances, whereas increases in pedestrian and bicycle activity may introduce offsetting effects through greater street-level activity and potential interactions at intersections.

Main Effects on EMS Response Times

We next turn to our main analyses of how the congestion policy, and resulting changes in traffic and mobility, translate into EMS response performance. Table 5 reports estimates for travel time to the incident scene, travel time from incident to hospital, and total EMS travel time, based on Equation (1), with the corresponding DD-RD plots presented in Figure 3. Panel A reports estimates based on all EMS incidents in Manhattan, while Panel B estimates are based on a restricted bandwidth of five kilometers on either side of the boundary. Both sets of estimates are broadly consistent, and point to a marked improvement in EMS performance within the congestion zone relative to nearby untreated areas following the introduction of congestion pricing.

Travel time to the incident declines by approximately 7–9 seconds (about 2 percent relative to the baseline mean; Panel B), though these estimates are imprecise. Travel time from incident to hospital falls by a larger margin of roughly 54–59 seconds (approximately 8 percent). Combining these two margins, total EMS travel time decreases by about 63–70 seconds (5–6 percent).⁹ These latter effects are precisely estimated and robust across alternate methods for measuring distance,

⁹That the relative magnitude of the EMS improvements is smaller than the underlying changes in traffic volumes reflects the fact that ambulances already operate with priority (e.g., lights and sirens, right-of-way, higher speeds) and therefore are less exposed to marginal congestion than standard vehicles. Moreover, increases in pedestrian and bicycle activity may partially offset these gains by increasing intersection density and street-level interactions.

Table 4: C2Smart Traffic Cameras: Difference-in-Discontinuities Estimates, Weekday vs. Weekend

	Weekday	Weekend
Passenger Vehicle		
DD-RD Estimate	-0.6868*** (0.0268)	-1.0185*** (0.0408)
Effect (%)	-18.1%	-26.8%
Pre-treatment Mean	3.8008	3.8008
Observations	1,674,413	686,200
Truck		
DD-RD Estimate	-0.0652*** (0.0051)	-0.1101*** (0.0067)
Effect (%)	-15.1%	-25.6%
Pre-treatment Mean	0.4308	0.4308
Observations	1,674,413	686,200
Pedestrian		
DD-RD Estimate	0.4803*** (0.0228)	0.2464*** (0.0353)
Effect (%)	15.6%	8.0%
Pre-treatment Mean	3.0752	3.0752
Observations	1,674,413	686,200
Bike		
DD-RD Estimate	0.0010** (0.0005)	0.0013* (0.0006)
Effect (%)	18.5%	23.3%
Pre-treatment Mean	0.0054	0.0054
Observations	1,674,413	686,200

Notes: DD-RD estimates by weekday/weekend (SE in parentheses), clustered on sensor.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 5: EMS Response: DD-RD Estimates ($D \times \text{Post}$), Alternate Distance-to-Boundary Measures

	Geometric Centroid	Pop.-Weighted Centroid	Uniform Random	Pop.-Weighted Random
Panel A: Full Sample (All Manhattan, No Bandwidth Restriction)				
<i>Time to Incident (sec)</i>				
DD-RD Estimate	-5.931 (4.979)	-6.277 (4.949)	-6.163 (5.001)	-5.585 (4.953)
Effect (%)	-1.2%	-1.3%	-1.3%	-1.2%
Pre-treatment Mean	484.24	484.24	485.10	485.08
Observations	496,390	496,390	486,169	487,325
<i>Time to Hospital (sec)</i>				
DD-RD Estimate	-24.202** (10.584)	-24.945** (10.374)	-22.726** (10.497)	-23.652** (10.360)
Effect (%)	-3.4%	-3.5%	-3.2%	-3.3%
Pre-treatment Mean	717.56	717.56	718.74	718.72
Observations	314,260	314,260	308,209	308,990
<i>Total Travel Time (sec)</i>				
DD-RD Estimate	-29.739** (13.503)	-31.142** (13.183)	-27.895** (13.562)	-28.821** (13.342)
Effect (%)	-2.5%	-2.6%	-2.3%	-2.4%
Pre-treatment Mean	1195.61	1195.61	1197.52	1197.43
Observations	311,519	311,519	305,495	306,272
Panel B: BW ≤ 5 km				
<i>Time to Incident (sec)</i>				
DD-RD Estimate	-9.039 (5.948)	-6.581 (5.770)	-7.714 (5.944)	-7.900 (5.699)
Effect (%)	-1.9%	-1.4%	-1.6%	-1.7%
Pre-treatment Mean	479.60	473.81	477.53	477.56
Observations	320,185	306,329	302,516	302,722
<i>Time to Hospital (sec)</i>				
DD-RD Estimate	-58.828*** (14.605)	-57.391*** (13.962)	-54.229*** (13.976)	-54.741*** (13.932)
Effect (%)	-8.2%	-8.1%	-7.6%	-7.7%
Pre-treatment Mean	715.34	705.37	713.14	712.32
Observations	201,221	191,567	189,815	189,923
<i>Total Travel Time (sec)</i>				
DD-RD Estimate	-70.427*** (19.137)	-63.547*** (17.608)	-63.279*** (18.740)	-63.239*** (17.971)
Effect (%)	-5.9%	-5.4%	-5.3%	-5.4%
Pre-treatment Mean	1188.09	1170.79	1183.04	1182.00
Observations	199,474	189,943	188,140	188,256

Notes: $D \times \text{Post}$ coefficient from the diff-in-discontinuities spec (outcome on D , running variable X , $D \times X$, Post , $X \times \text{Post}$, $D \times X \times \text{Post}$, and $D \times \text{Post}$, with FEs for year-month, hour, day-of-week, initial/final severity, initial/final call type, disposition, and dispatch area); SE in parentheses, clustered on intersection polygon (`poly_zone_id`). Columns differ only in the distance-to-boundary measure used as the running variable X : *Geometric Centroid* and *Pop.-Weighted Centroid* use the (population-weighted) centroid of each intersection polygon; *Uniform Random* and *Pop.-Weighted Random* use a (population-weighted) randomly sampled point within each polygon. Panel B restricts each column's sample to incidents within 5 km of the boundary, using that column's own distance measure. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

and demonstrate that congestion pricing yields meaningful improvements across the full EMS response cycle.¹⁰

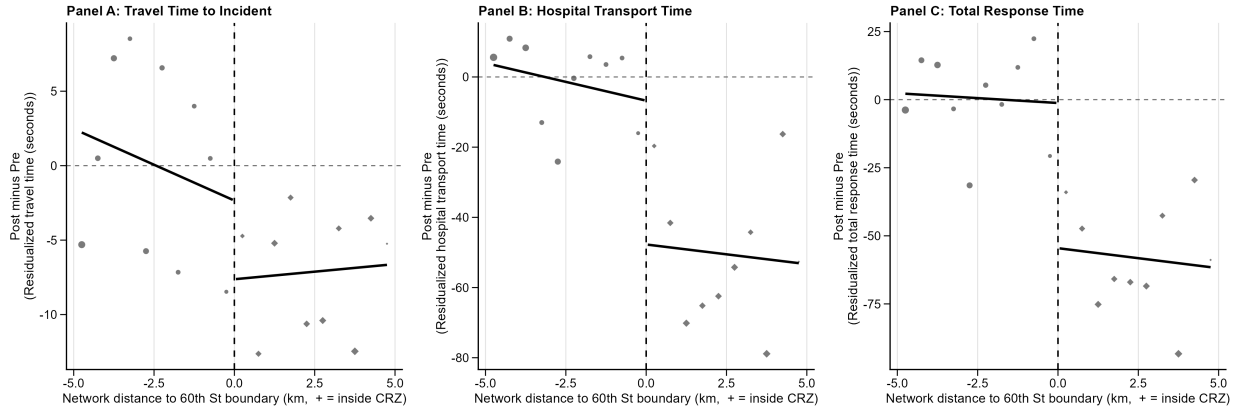


Figure 4: Difference-in-Discontinuities in EMS Times

These findings align closely with the first-stage evidence documented in Tables 3 and 4. The observed reductions in passenger and truck traffic within the congestion zone provide a direct mechanism through which congestion pricing alleviates delays faced by emergency vehicles.

To further gauge the plausibility of the estimated magnitudes, we can conduct a simple back-of-the-envelope calculation based on observed improvements in traffic speeds following the implementation of congestion pricing. The average resident in midtown Manhattan is typically located approximately 0.5 to 1.5 miles from the nearest hospital or emergency department (see Appendix Figure A2), and pre-policy traffic speeds south of 60th Street averaged approximately 6.9 to 8.2 miles per hour (Cook et al., 2025; Hoylman-Sigal and Schwartz, 2024).¹¹

Cook et al. (2025) estimate that average vehicle speeds increased by roughly 1.5 miles per hour (about 15 percent) following congestion pricing. Under the simplifying assumption that ambulance travel times scale proportionally with average traffic speeds, these changes imply potential reductions in hospital transport time on the order of 34–47 seconds for a 0.5-mile trip and 102–140 seconds for a 1.5-mile trip. While these calculations are necessarily approximate and rely on a stylized relationship between speed and travel time, they provide a useful benchmark for interpreting our empirical estimates.

Importantly, this exercise abstracts from other channels through which congestion pricing may affect EMS performance, including increased pedestrian and bicycle activity, intersection delays, and operational or behavioral responses within the EMS system. Nevertheless, the fact that our estimated reductions in hospital transport times fall well within the range implied by observed increases in traffic speeds provides external validation for the magnitude of our findings and reinforces the credibility of the DD-RD estimates.

¹⁰These findings are also largely robust to more restricted bandwidths and alternate functional forms (quadratic trends, local linear regression); see A1.

¹¹Note that these figures (distance to the nearest hospital and mean speeds) are consistent with the observed mean travel time to the hospital, of 11–12 minutes, for the average EMS incident in Manhattan (Table 1).

Table 6 examines heterogeneity in the EMS response effects across weekday versus weekend periods and peak versus off-peak hours. The estimates show that overall improvements in EMS performance are present throughout the week and across all hours, though the pattern points to somewhat larger improvements on weekdays compared to weekends (6 percent versus 4 percent). For travel time to the incident, meaningful reductions are evident only on weekdays and during peak hours (11–12 seconds improvement; approximately 2 percent), whereas for transport time from incident to hospital, reductions are significant across both weekday and weekend periods, with some evidence of larger improvements on weekdays.

Table 6: EMS Response - $D \times \text{Post}$ Estimates, Timing Heterogeneity

	Panel A: Day Type		Panel B: Time of Day	
	Weekday	Weekend	Peak	Off-Peak
Time to Incident (sec)				
$D \times \text{Post}$	-11.715*	0.588	-11.386	0.515
	(6.707)	(6.613)	(7.068)	(8.322)
Effect (%)	-2.4%	0.1%	-2.3%	0.1%
Pre-treatment Mean	482.73	464.28	495.80	430.10
Observations	217,647	85,075	219,118	83,604
Time to Hospital (sec)				
$D \times \text{Post}$	-59.045***	-44.222***	-56.158***	-56.786***
	(14.538)	(15.765)	(15.226)	(14.101)
Effect (%)	-8.2%	-6.4%	-7.5%	-9.2%
Pre-treatment Mean	722.02	686.38	748.63	619.15
Observations	138,089	51,834	136,875	53,048
Total Travel Time (sec)				
$D \times \text{Post}$	-70.024***	-47.194**	-66.239***	-62.995**
	(19.961)	(21.858)	(19.470)	(24.357)
Effect (%)	-5.9%	-4.1%	-5.4%	-6.0%
Pre-treatment Mean	1196.55	1143.23	1236.17	1044.27
Observations	136,748	51,508	135,341	52,915

Notes: $D \times \text{Post}$ coefficient (SE in parentheses), as in Table 4. Peak = weekdays 5am-9pm and weekends 9am-9pm; Off-Peak = all other hours. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Relating these findings to the first-stage results in Table 4 provides additional context. While vehicular density reductions are somewhat larger on weekends, the EMS response improvements do not map one-for-one into larger weekend effects. This likely reflects offsetting differences in other dimensions of mobility. Increases in pedestrian activity are more muted on weekends, while increases in bicycle density are more pronounced. These shifts may differentially affect ambulance speed, for example, higher levels of pedestrian and cyclist activity may increase intersection frictions, partially offsetting gains from reduced vehicular congestion. Additionally, the marginal benefits of congestion relief for emergency vehicle movement may be greatest during periods of higher baseline

congestion, such as weekdays and peak hours, when even modest reductions in traffic can translate into improvements in travel times.

Identification and Robustness Checks

We conduct a series of robustness and placebo exercises to assess the validity of the empirical design. First, we implement a placebo test using only pre-policy data (2023–2024), assigning a pseudo-treatment date of January 5, 2024, and estimating the DD-RD specification. As shown in Table 7, we find no evidence of discontinuous changes in EMS response times at the boundary under this placebo assignment. All estimated effects are small in magnitude and statistically indistinguishable from zero across travel time to the incident, transport time to the hospital, and total travel time. These results support the plausibility of the identifying assumption that, absent the policy, outcomes would have evolved smoothly across the boundary.

Table 7: EMS Response - Placebo Checks (Pseudo-Treatment and Falsification)

	Pseudo-Treatment (2023–2024 Data, Pseudo-Post = Jan. 2024)			Falsification: On-Scene Time (Real Post)	
	Time to Incident	Time to Hospital	Total Travel Time	No Hosp. Transport	Non-severe & No Transport
D × Pseudo-Post / Post	5.960 (4.711)	-11.118 (9.223)	-0.379 (11.769)	-2.245 (27.796)	-4.575 (32.527)
Effect (%)	1.3%	-1.7%	-0.0%	-0.1%	-0.3%
Pre-treatment Mean	457.54	668.97	1115.81	1808.31	1758.02
Observations	336,242	209,731	207,788	112,816	63,817

Notes: SE in parentheses, clustered on `poly_zone_id` (intersection polygon), restricted to $BW \leq 5$ km, as in Table 4. Columns 1–3 report D × Pseudo-Post from the pseudo-treatment placebo (2023–2024 data, placebo date Jan. 5, 2024); columns 4–5 report D × Post (real CRZ start, Jan. 5, 2025) from the on-scene (non-travel) time falsification outcome. “No Hosp. Transport” restricts to `INCIDENT_DISPOSITION_CODE` $\neq$ 82; “Non-severe & No Transport” additionally restricts to `INITIAL_SEVERITY_LEVEL_CODE` 4–8. Both placebo checks should be null/insignificant. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Second, we examine outcomes that should not be affected by congestion pricing. Specifically, we analyze time spent at the incident scene for cases that do not involve hospital transport and for a subset of non-severe incidents without transport. As expected, these estimates are also small and statistically insignificant, indicating no meaningful effects of congestion pricing on non-travel components of EMS response. Taken together, these placebo and falsification tests provide support for the validity of the research design and reinforce the interpretation that the estimated improvements in EMS response times reflect changes in travel conditions rather than spurious discontinuities or shifts in incident characteristics.

Finally, we estimate the DD-RD model for a set of incident-level characteristics and system outcomes that should remain smooth across the boundary after differencing out any baseline discontinuities, including the total number of EMS incidents, incident severity, call type, and disposition. As shown in Table 8, we find no evidence of economically meaningful discontinuities in these variables at the boundary following the implementation of congestion pricing. While a small number of coefficients—particularly for severity—are statistically significant, their magnitudes are extremely small relative to baseline means and do not indicate substantively important changes in the composition or nature of EMS incidents.

One potential concern is that contemporaneous changes in transportation infrastructure could

Table 8: Identification Checks: $D \times \text{Post}$ on Pre-Treatment Determinants

	$D \times \text{Post}$	(SE)	Pre-treatment Mean	Observations
Incident counts (zone $\times$ year-month panel)	-6.466	(5.029)	126.242	2,542
Severity (continuous, 1-9)	0.013***	(0.004)	3.855	305,829
Severe incident (severity 1-3 vs. 4-8)	0.002**	(0.001)	0.478	305,829
Cardiac/respiratory call type	-0.006	(0.006)	0.205	305,829
Hospital transport (disposition = 82)	0.003	(0.006)	0.634	305,829

Notes: $D \times \text{Post}$ coefficient (SE in parentheses), as in Table 4. Row 1 is a zone $\times$ year-month panel of incident counts; rows 2-5 use the corresponding variable as the outcome in place of its usual fixed effect. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

affect local traffic conditions and therefore bias the estimated effects. In particular, bicycle lane expansions and roadway construction activity may have been differentially distributed across areas inside and outside the congestion relief zone during the study period. To assess this possibility, Table A2 augments the baseline DD-RD specification with controls for bicycle lane infrastructure, recent bicycle facility upgrades, and zone-by-month measures of active roadway construction activity derived from NYC Department of Transportation (DOT) permit records.

The estimated effects remain highly similar to the baseline results after including these additional controls. The stability of the estimates across specifications suggests that the results are not driven by concurrent changes in bicycle infrastructure or roadway construction that may have differentially affected local traffic conditions on either side of the boundary.

Taken together, these findings suggest that the congestion pricing policy did not induce sorting or discontinuous changes in incident characteristics around the boundary. This provides additional support for the validity of the research design and reinforces the interpretation that the estimated improvements in EMS response times are not driven by compositional changes in incidents or other spurious discontinuities.

Response Dynamics

We next examine the dynamics of the treatment effects using cumulative DD-RD specifications that allow the discontinuity at the boundary to evolve over time following the implementation of congestion pricing. Figure 5 presents estimates for EMS response outcomes, where each point reflects the cumulative effect using data through month k of 2025. The estimates indicate that improvements in EMS performance emerge quickly following the introduction of the policy. In particular, reductions in incident response time, hospital transport time, and total travel time are evident within the first few months post-implementation, with economically meaningful declines relative to the pre-policy baseline.

As the post-policy window expands, the magnitude of these improvements evolves. Rather than exhibiting a simple monotonic attenuation, the estimates suggest a gradual stabilization of effects, with the largest gains observed early on and smaller—but still negative—effects persisting over a longer horizon. This pattern indicates that congestion pricing generates immediate improvements

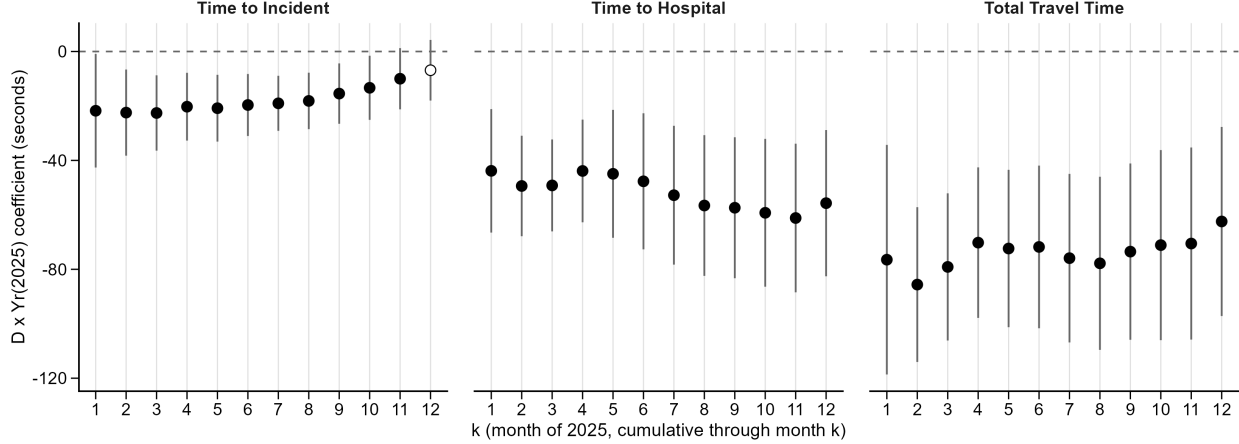


Figure 5: Cumulative Impact on EMS Times

in emergency response conditions, followed by a transition toward a new equilibrium in which gains are partially maintained over the observed horizon rather than fully dissipating.

A similar dynamic pattern is evident in the first-stage results for traffic and mobility (Figure 6), where treatment effects are estimated cumulatively over progressively longer post-policy windows (2, 4, 6, and 9 weeks).¹² Reductions in passenger vehicle and truck traffic materialize and become more pronounced in the initial weeks following implementation before stabilizing as the sample window expands. At the same time, increases in pedestrian and bicycle activity emerge quickly and remain relatively stable over time. The temporal alignment between the stabilization in vehicular traffic reductions and the stabilization of EMS improvements reinforces the interpretation that changes in traffic conditions are the primary mechanism linking congestion pricing to EMS response outcomes.

Taken together, these results suggest that congestion pricing induces sharp short-run adjustments in mobility—particularly reductions in vehicle traffic—that translate into immediate improvements in EMS response performance. Over time, behavioral responses such as route substitution, trip rescheduling, and adaptation to the new pricing regime appear to mildly offset these initial gains, leading to a smaller but persistent improvement in travel times. This pattern is consistent with a transition from short-run responses to a medium-run equilibrium, in which congestion pricing continues to deliver benefits for emergency services, albeit at a reduced magnitude relative to the initial post-policy period. This pattern is also consistent with prior evidence showing that congestion pricing generates strong initial responses that evolve over time as travelers adjust their behavior.

Spatial Patterns and Displacement

We next examine spatial heterogeneity and potential displacement of traffic and EMS effects using the spatial event-study framework, Equation 2), which traces treatment effects across 500-meter

¹²Note that since the C2Smart traffic sensor data are only available for approximately the first nine weeks post-policy, we are not able to assess more extended dynamics in the first-stage responses as we are with the EMS responses.

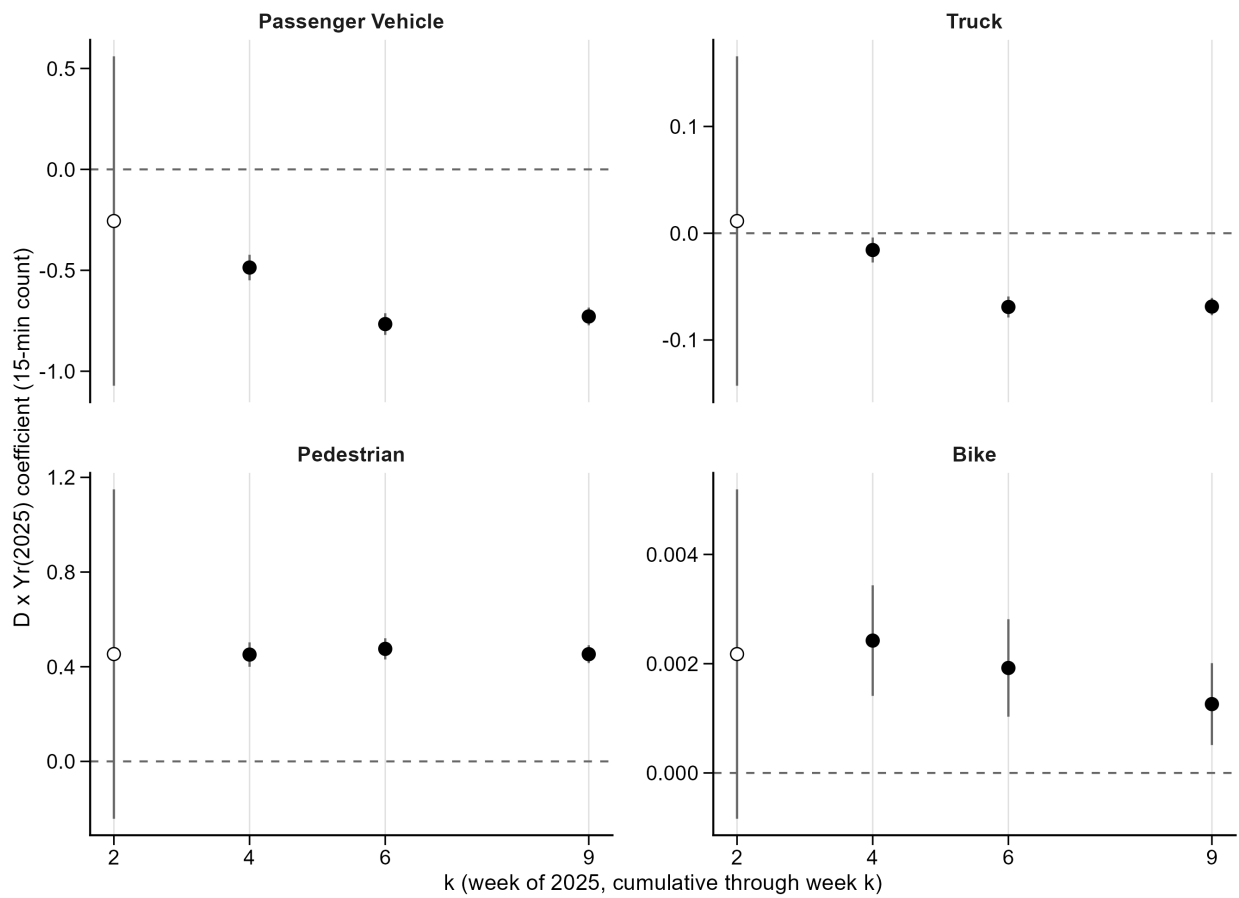


Figure 6: First Stage Dynamics

distance bands near the boundary (up to ± 2 km) and wider (1 km) bands farther away, extending up to 5 kilometers on either side. Figures 7 and 8 present bin-specific DD-RD estimates along with separate local polynomial (locally-estimated scatterplot smoothing; LOESS) smoothers on each side of the boundary to highlight underlying spatial patterns while preserving the discontinuity at the cutoff.

Figure 7 presents first-stage results for traffic and mobility outcomes. Reductions in passenger vehicle and truck traffic are concentrated within the congestion zone and in close proximity to the 60th Street boundary, with effects attenuating rapidly as distance from the boundary increases. There is some evidence of modest increases in vehicular traffic immediately north of the boundary, consistent with limited rerouting or toll-avoidance behavior. However, these increases are small in magnitude, geographically localized, and imprecisely estimated beyond the nearest bands. Overall, the spatial pattern suggests that congestion pricing primarily reduces traffic within the zone rather than inducing large-scale displacement to surrounding areas.

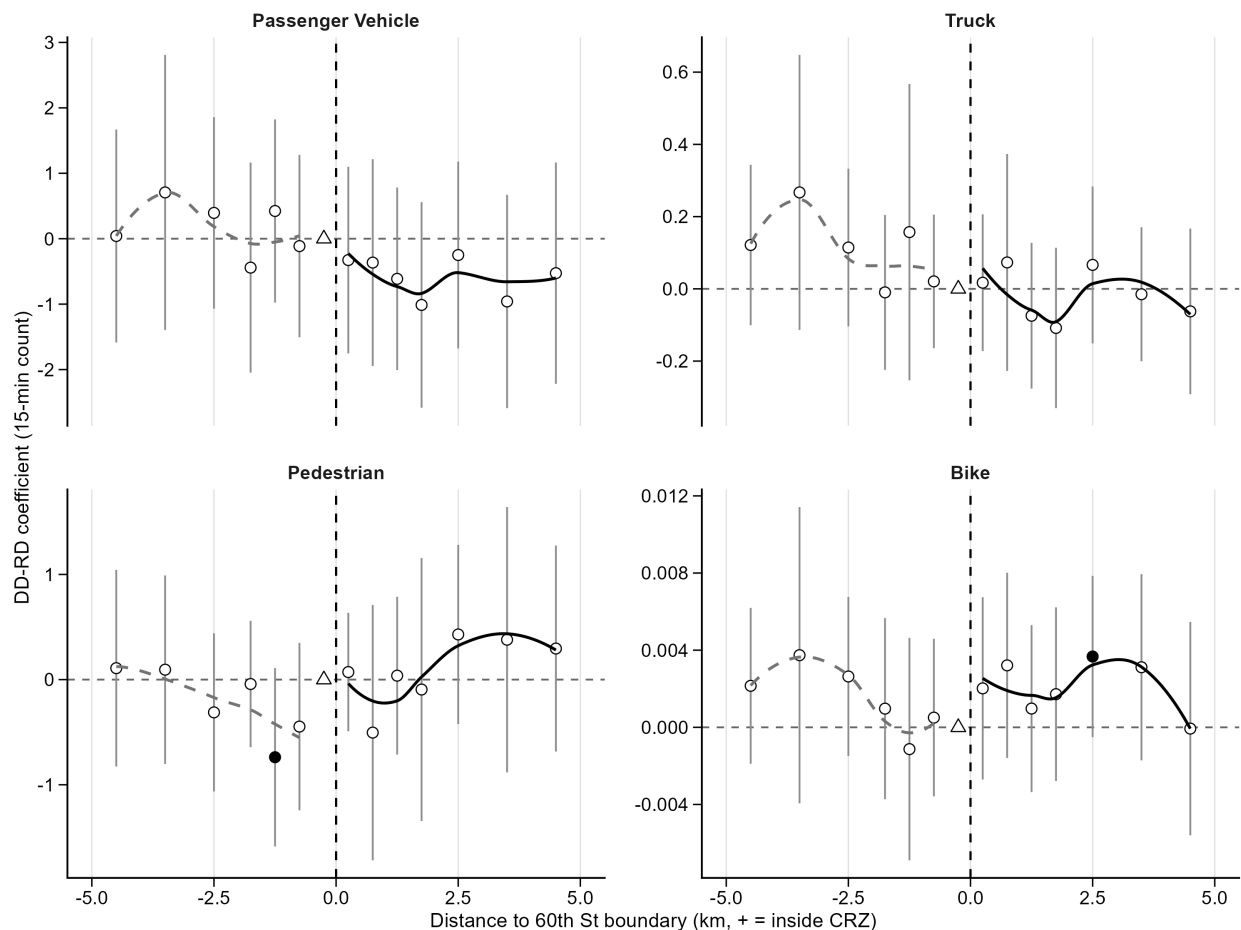


Figure 7

Turning to EMS outcomes, shown on Figure 8), we observe a broadly similar spatial pattern. Improvements in response times are concentrated near the boundary inside the congestion zone, with reductions in both incident response time and hospital transport time that are largest within

roughly one to two kilometers of the cutoff. As distance from the boundary increases, these effects attenuate and become less precisely estimated; in more distant bands within the congestion zone, coefficients approach zero and in some cases modestly exceed it. Importantly, however, there is little systematic evidence of worsening EMS response times immediately outside the congestion zone, despite the modest increases in traffic observed in those areas. Taken together, these findings suggest that any displacement of traffic is limited in scope and does not translate into meaningful deterioration in EMS performance in nearby untreated neighborhoods.

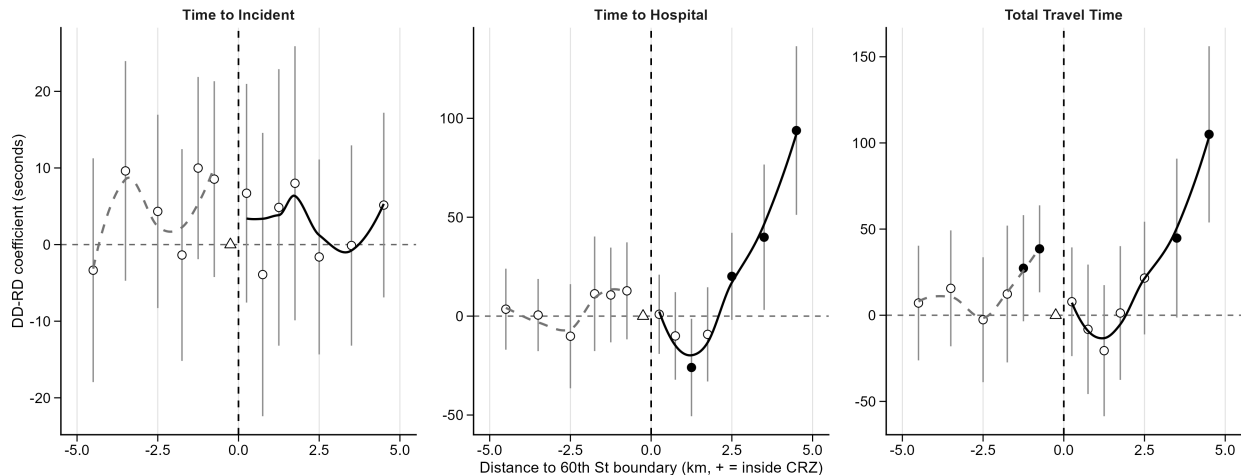


Figure 8

The smoothed spatial profiles highlight a common pattern across outcomes: sharp reductions in traffic and corresponding improvements in EMS response times near the policy boundary, followed by rapid attenuation with distance. This pattern is consistent with congestion pricing primarily affecting marginal inflows into the congestion zone rather than uniformly improving conditions throughout Manhattan.

One notable feature is that, for hospital transport time and total EMS travel time, some distance bands deeper within the congestion zone (approximately 2.5 to 5 kilometers south of the boundary) exhibit positive coefficients. These patterns should not be interpreted as evidence of displacement, which would instead manifest as worsening outcomes immediately outside the congestion zone. Rather, they reflect heterogeneity in treatment effects within the zone.

Because the DD-RD design identifies local causal effects at the boundary, estimates farther from the cutoff increasingly reflect a combination of treatment exposure and broader equilibrium dynamics within the urban network. In particular, ambulance transport times depend on factors such as hospital location, network congestion, and interactions with pedestrian and bicycle traffic, which are especially pronounced in central Manhattan. As a result, reductions in vehicular inflows at the boundary need not translate uniformly into improved travel times deeper within the urban core.

The spatial patterns also complement the dynamic evidence presented earlier. The concentration of both traffic reductions and EMS improvements near the boundary is consistent with the

temporal results showing that effects are strongest in the short run, when reductions in vehicular inflows are most pronounced. As drivers adjust their behavior over time and traffic conditions partially equilibrate, the marginal benefits of congestion reduction appear to diminish—mirroring both the temporal attenuation observed in the first-stage estimates and the spatial attenuation seen with distance from the boundary.

Taken together, these findings suggest that congestion pricing generates localized improvements in traffic conditions that translate into measurable gains in EMS response performance near the policy boundary, with effects that attenuate across both space and time. While there is some heterogeneity within the congestion zone, there is little evidence of systematic spillovers just outside the boundary. As such, the results provide little to no evidence that congestion pricing simply displaces congestion to nearby areas, and instead indicate policy effects that are consistent with net improvements in mobility conditions relevant for emergency services.

Difference-in-Differences Estimates

As a point of comparison, Table 9 reports estimates from a standard difference-in-differences (DD) specification, based on Equation (3) that compares outcomes inside and outside the congestion pricing zone before and after policy implementation. In contrast to the DD-RD results, the DD estimates show small and statistically insignificant reductions in travel time to the incident, but positive and statistically significant increases in transport time to the hospital and total EMS travel time. In particular, hospital transport times increase by approximately 18–34 seconds (2.6–4.8 percent), and total travel time increases by approximately 17–33 seconds (1.4–2.7 percent), depending on the sample.

To better understand these patterns, Figure 9 presents DD event-study estimates that trace the evolution of effects over time. The pre-policy coefficients are generally stable and close to zero, providing some support for parallel trends. However, the post-policy dynamics reveal a more nuanced pattern: while there is little evidence of immediate increases in travel times following the introduction of congestion pricing in January 2025, transport times begin to rise more noticeably in subsequent periods. Notably, this timing closely coincides with the implementation of the FDNY directive on March 12, 2025, which required ambulances to transport patients to the nearest hospital rather than following previous routing protocols.¹³

To isolate the role of this concurrent policy change, Table 10 augments the DD specification by including an interaction between the post-directive period and the treatment indicator. The results reveal a clear reversal in the estimated effects. Prior to the directive, congestion pricing is associated with significant reductions in travel times across all three outcomes, with decreases of roughly 20–50 seconds (2–5 percent) depending on the outcome. However, these gains are offset—and in some cases reversed—after the implementation of the directive, as the interaction term shows large positive and statistically significant effects on hospital transport time and total travel time.

¹³ A study in five western regions in the U.S. found that patient/family choice played a determining role in hospital selection in over half (50.6 percent) of the EMS calls (Newgard et al. 2013).

Table 9: EMS Response: Difference-in-Differences Estimates ($D \times \text{Post}$)

	Full Sample	BW ≤ 5 km
Time to Incident (sec)		
$D \times \text{Post}$	-3.684 (2.515)	-2.040 (2.939)
Effect (%)	-0.8%	-0.4%
Pre-treatment Mean	485.08	477.56
Observations	487,325	302,722
Time to Hospital (sec)		
$D \times \text{Post}$	34.344*** (8.448)	18.228 (11.283)
Effect (%)	4.8%	2.6%
Pre-treatment Mean	718.72	712.32
Observations	308,990	189,923
Total Travel Time (sec)		
$D \times \text{Post}$	32.803*** (9.513)	16.848 (12.370)
Effect (%)	2.7%	1.4%
Pre-treatment Mean	1197.43	1182.00
Observations	306,272	188,256

Notes: $D \times \text{Post}$ coefficient from a standard DiD spec (SE in parentheses, clustered on intersection polygon). Shown for comparison with the DD-RD estimates in Table 4. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

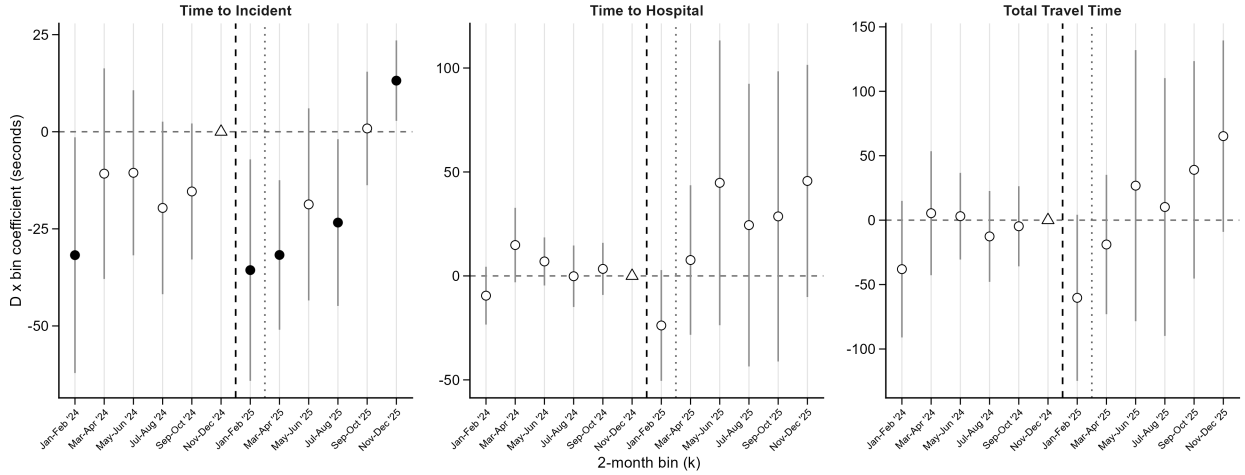


Figure 9: Event Study - EMS Times

As a result, the overall DD estimates, which pool across both periods, reflect a combination of these opposing forces.

We interpret these findings as reflecting the limitations of the standard DD design in this setting. In particular, the DD approach compares outcomes across geographically distinct areas that may experience differential time trends and may be differentially affected by system-wide operational changes such as the FDNY directive. Because the directive directly affects hospital transport behavior, and plausibly varies across locations, it introduces a confounding influence that is not easily separated from the effect of congestion pricing in a standard DD framework.

In contrast, the difference-in-discontinuities design focuses on localized comparisons at the 60th Street boundary and differences out any pre-existing discontinuities and common time trends. As a result, it is less sensitive to citywide policy changes that affect both sides of the boundary similarly, as reflected in the relatively stable dynamics pre- and post-FDNY directive (Figure 5). The fact that the pre-directive DD estimates align with the DD-RD findings—showing reductions in EMS response times—provides additional evidence consistent with this interpretation.

Conclusion

This paper provides the first quasi-experimental evidence that congestion pricing generates a novel and previously unquantified external benefit: faster emergency medical service response times. Exploiting the sharp geographic boundary of New York City’s congestion relief zone at 60th Street in Manhattan and a difference-in-discontinuities design, we find that the January 2025 implementation of congestion pricing produced substantial improvements in EMS performance within the congestion price zone.

First-stage results confirm that the policy generated substantial, localized reductions in vehicular traffic near the boundary, passenger vehicle density fell by roughly 21 percent and truck traffic by 18 percent, accompanied by corresponding increases in pedestrian and bicycle activity. These mobility changes provide evidence of a direct mechanism through which congestion pricing plausibly affects ambulance travel times.

The main EMS results indicate that travel time to incident scenes declined by approximately 7–9 seconds, hospital transport time fell by roughly 54–59 seconds, and total travel time decreased by 63–70 seconds, a reduction of approximately 5–6 percent relative to pre-treatment times. These effects are precisely estimated, robust to alternative distance measures and functional forms, and survive a battery of placebo and falsification tests. Dynamic estimates indicate that improvements emerged quickly following implementation and, while partially attenuating over subsequent months as drivers adjusted their behavior, remained negative throughout the observed post-policy window. Spatial analysis further confirms that these gains are concentrated near the policy boundary and do not appear to be driven by congestion displacement to adjacent areas. However, we note that our estimates capture localized effects near the policy boundary and may not fully reflect broader general equilibrium adjustments across the entire city.

Table 10: EMS Response - DiD with Post-Directive Interaction ($D \times \text{Post}$ and $D \times \text{Post-Directive}$)

	Full Sample	BW ≤ 5 km
Time to Incident (sec)		
$D \times \text{Post}$ (pre-directive)	-23.161*** (4.396)	-21.751*** (4.231)
$D \times \text{Post-Directive}$ (incremental)	23.522*** (4.390)	24.645*** (5.894)
Total Effect ($\geq \text{Mar. 12}$)	0.361 (2.847)	2.894 (3.286)
Effect (% , pre-directive)	-4.8%	-4.6%
Effect (% , total, $\geq \text{Mar. 12}$)	0.1%	0.6%
Pre-treatment Mean	485.08	477.56
Observations	487,325	302,722
Time to Hospital (sec)		
$D \times \text{Post}$ (pre-directive)	-13.416 (10.334)	-27.617** (9.918)
$D \times \text{Post-Directive}$ (incremental)	61.398*** (14.476)	61.414** (18.603)
Total Effect ($\geq \text{Mar. 12}$)	47.982* (22.800)	33.797 (27.272)
Effect (% , pre-directive)	-1.9%	-3.9%
Effect (% , total, $\geq \text{Mar. 12}$)	6.7%	4.7%
Pre-treatment Mean	718.72	712.32
Observations	308,990	189,923
Total Travel Time (sec)		
$D \times \text{Post}$ (pre-directive)	-34.980** (14.600)	-50.081** (15.235)
$D \times \text{Post-Directive}$ (incremental)	84.864*** (11.729)	86.803*** (17.060)
Total Effect ($\geq \text{Mar. 12}$)	49.884* (22.477)	36.722 (27.203)
Effect (% , pre-directive)	-2.9%	-4.2%
Effect (% , total, $\geq \text{Mar. 12}$)	4.2%	3.1%
Pre-treatment Mean	1197.43	1182.00
Observations	306,272	188,256

Notes: Estimates from outcome $\sim D + \text{Post} + D \times \text{Post} + \text{Post-Directive} + D \times \text{Post-Directive} \mid \text{FE_POLY (year} \times \text{month, hour, day-of-week, severity, call type, disposition, dispatch area, and 195-zone intersection-polygon FE; D absorbed by poly_zone_id), SE clustered on dispatch area. “}D \times \text{Post (pre-directive)}” \text{ is differential CRZ effect Jan. 5–Mar. 11, 2025; “}D \times \text{Post-Directive (incremental)}” \text{ is additional effect from Mar. 12, 2025 onward; “Total Effect (} \geq \text{Mar. 12)}” \text{ is their sum. ***}p < 0.01; **p < 0.05; *p < 0.10.$

A notable complication in the analysis is the FDNY directive issued in March 2025 requiring ambulances to transport patients to the nearest hospital rather than a preferred facility. This concurrent operational change substantially altered hospital transport patterns citywide, confounding standard difference-in-differences estimates. After accounting for this directive, the pre-directive DD estimates align closely with our main DD-RD findings, providing additional confidence in the causal interpretation of our results. This episode also illustrates that evaluations of urban transportation policies in complex institutional environments must carefully account for concurrent policy changes that can obscure or reverse treatment effects.

These findings carry several implications. For cost-benefit analyses of congestion pricing, standard frameworks that weigh implementation costs against commuter time savings and environmental gains are likely to understate net social benefits if EMS response-time improvements are omitted. Translating our estimated reductions in response time through the mortality elasticity documented by Wilde (2013), approximately one percentage point per additional minute of response time, suggests that even modest improvements in ambulance travel times may generate important welfare consequences, particularly given the volume of EMS incidents in a dense urban environment like Manhattan. Future work should seek to quantify these downstream health effects directly.

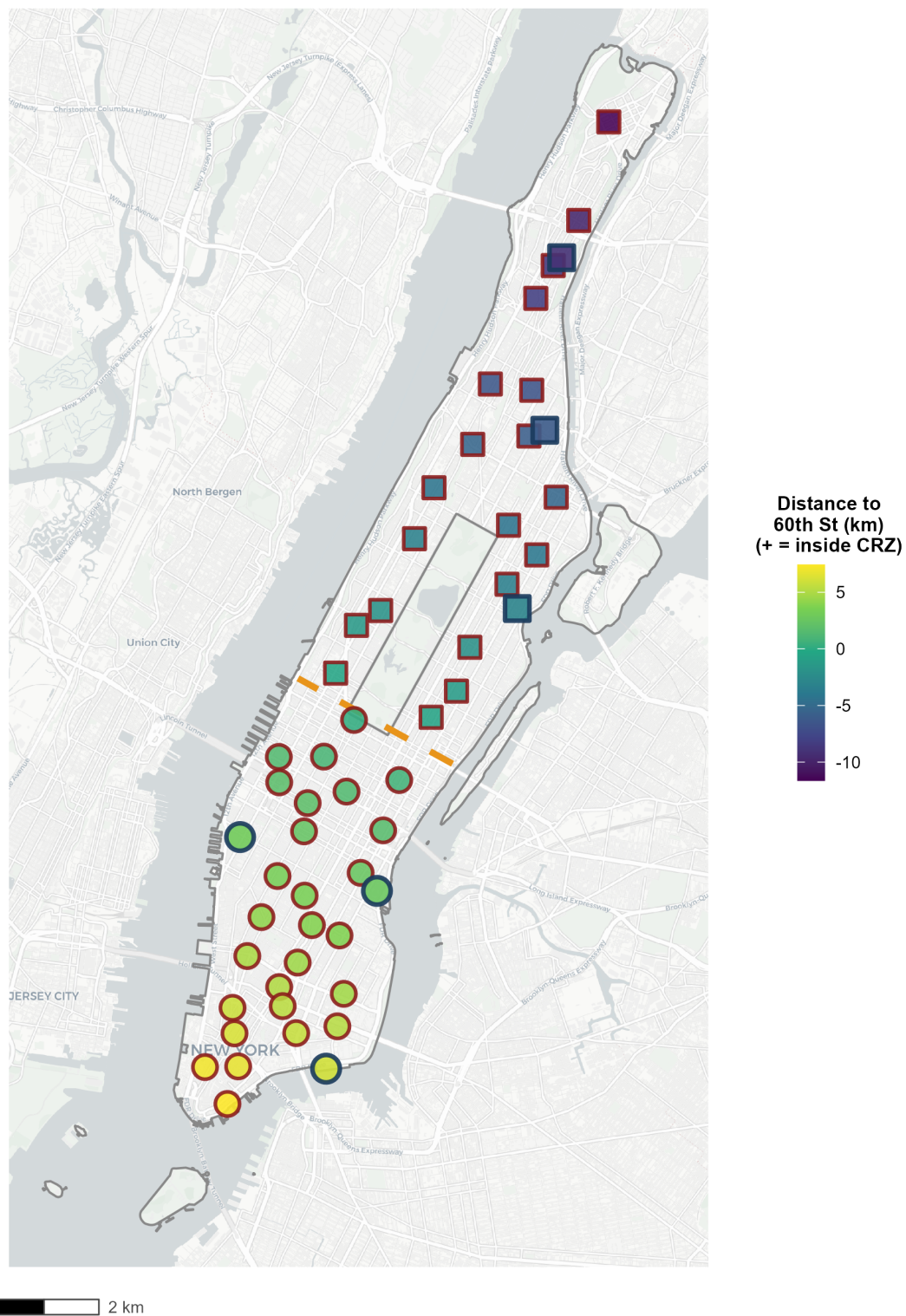
For urban transportation policy more broadly, our results suggest that emergency response system performance should be treated as a distinct welfare channel alongside the more commonly studied outcomes of commuter travel time, accident rates, and air quality. The complementarity between road congestion and public good provision documented here, that congestion pricing that reduces vehicle volumes also improves the speed and reliability of emergency services, points to a category of benefit that economists have only recently begun to model formally. As congestion pricing programs expand to additional cities, incorporating EMS response information into program evaluations will be essential for producing comprehensive assessments of their social value.

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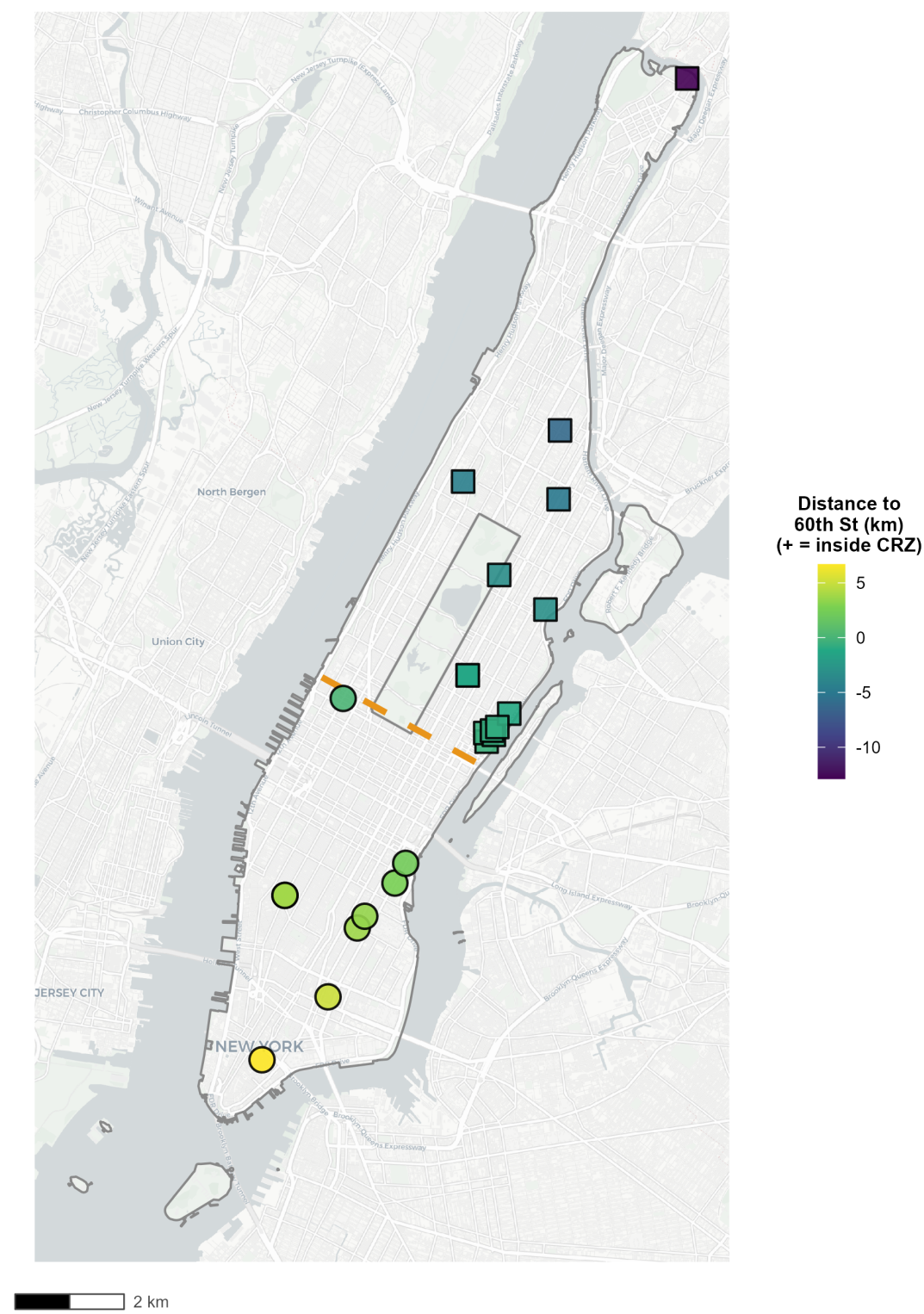
Shape: Circle = inside CRZ, Square = outside CRZ | Outline: navy = Ambulance (n=6), red = Fire (n=47)



(c) OpenStreetMap / CartoDB | Source: NYC Public Facility data.

Figure A1: EMS and Fire Station Locations

Circle = inside CRZ (n=9); Square = outside CRZ (n=14)



(c) OpenStreetMap / CartoDB | Source: NYS Health Facility data.

Figure A2: Hospital Distribution

Table A1: Appendix Table 1: EMS DD-RD Estimates Across Bandwidths and Functional Forms

	Time to Incident	Time to Hospital	Total Travel Time
Panel A: Bandwidth variations (linear)			
BW = 2 km	-7.237 (7.193)	-3.141 (15.510)	-26.981 (21.586)
BW = 3 km	-3.920 (6.775)	-43.180*** (14.170)	-55.934*** (20.024)
BW = 4 km	-3.099 (6.286)	-45.974*** (13.677)	-49.239** (19.225)
BW = 5 km (main)	-7.900 (5.699)	-54.741*** (13.932)	-63.239*** (17.971)
Full sample (BW = 10 km)	-3.473 (4.687)	-25.132** (10.961)	-26.911* (13.711)
Panel B: Functional form (BW = 5 km)			
Quadratic (BW = 5 km)	-1.988 (7.206)	-14.539 (14.038)	-29.858 (22.455)
Local linear (BW = 5 km)	-4.349 (5.853)	-44.026*** (13.094)	-53.972*** (17.845)
<i>Observations (BW = 5 km)</i>	302,722	189,923	188,256

Notes: Each cell shows the DD-RD coefficient on the D×Post interaction (route-based treatment) with standard errors in parentheses, clustered by intersection polygon. *** p<0.01, ** p<0.05, * p<0.10. Panel A varies the bandwidth (distance to 60th St boundary) using the linear functional form. Panel B uses BW = 5 km with a quadratic polynomial in distance or a local linear fit (triangular kernel weights). Fixed effects: year-by-month, hour-of-day, day-of-week, initial severity, disposition code, dispatch area.

Table A2: EMS DD-RD: Robustness to Bike Lane and Road Construction Controls

Outcome	Sample	Baseline		+Bike Lane Ctls		+Bike Lane + Construction	
		Coef.	SE	Coef.	SE	Coef.	SE
Time to Incident	Full sample	-5.585	(3.502)	-6.922*	(3.398)	-6.991	(4.079)
	BW $\leq$ 5km	-7.900**	(3.034)	-9.297**	(3.017)	-9.329**	(3.917)
Time to Hospital	Full sample	-23.652	(19.681)	-26.526	(19.685)	-26.562	(19.854)
	BW $\leq$ 5km	-54.741**	(16.631)	-55.267**	(16.781)	-55.622**	(18.370)
Total Travel Time	Full sample	-28.821	(20.824)	-32.755	(20.910)	-32.689	(22.372)
	BW $\leq$ 5km	-63.239***	(18.555)	-65.329***	(18.708)	-65.056**	(24.102)

Notes: Route-based treatment; raster distance running variable. Bike lane controls: cumulative on-street lane-miles (protected, conventional, shared) and a static zone-level flag for zones with any facility upgrade (retirement) in Jul–Dec 2024 (12 zones); NYC DOT Bike Routes data, on-street segments only. Construction control: roadway-permit-active-days per zone $\times$ month, NYC DOT Street Construction Permits, roadway-relevant types only. Year-month, hour, day-of-week, severity, call type, disposition, and dispatch-area FE. SE clustered by dispatch area. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.